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(54) **MULTIPOLE ELEMENT, IMAGE ERROR CORRECTOR AND PARTICLE BEAM SYSTEM**

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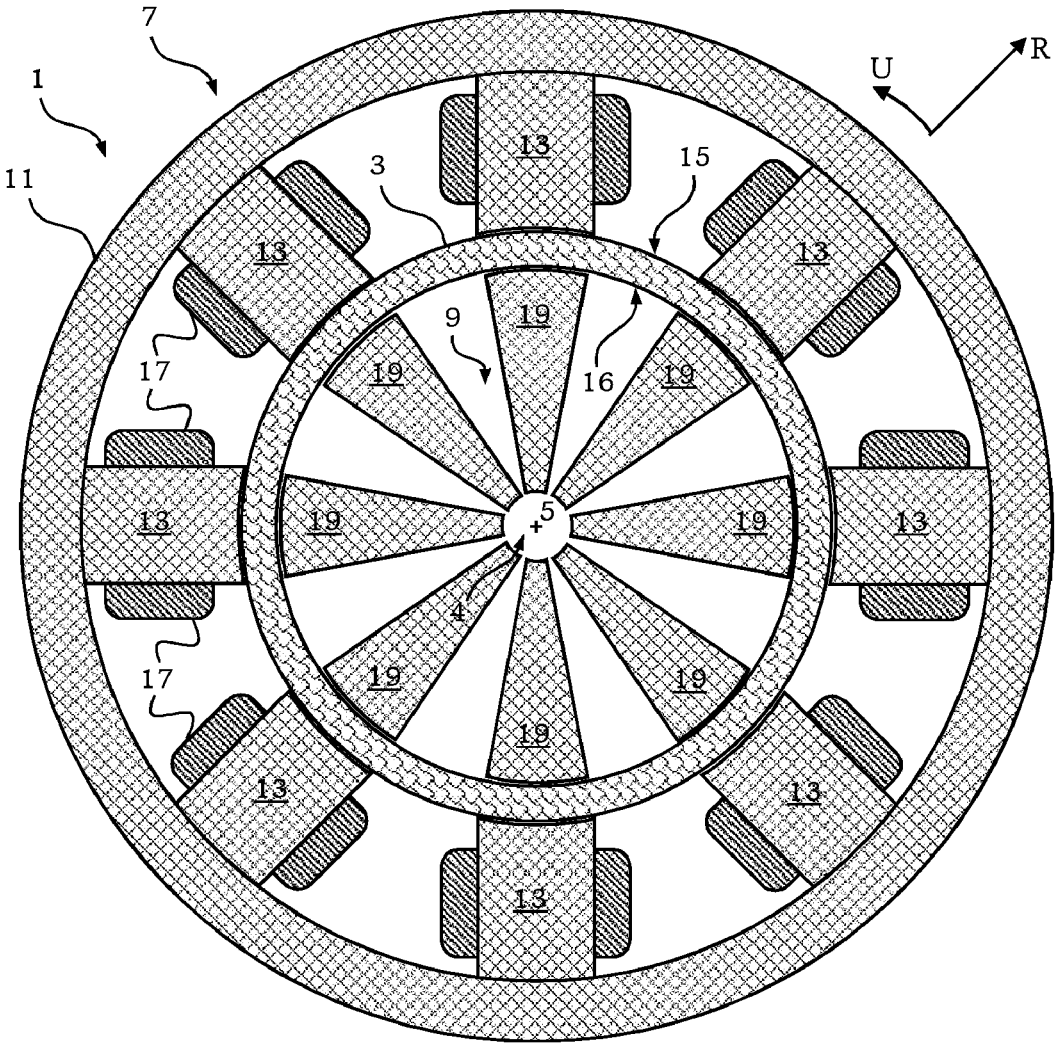
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CPC ***H01J 37/153*** (2013.01); ***H01J 37/141*** (2013.01); *H01J 2237/141* (2013.01); *H01J 2237/153* (2013.01)

(57) **ABSTRACT**

A multipole element for creating a magnetic multipole field or for creating an electric-magnetic multipole field for a particle beam system such as a scanning electron microscope, for example, comprises: a tube surrounding a central axis of the multipole element; an external space assembly arranged outside of the tube and a vacuum space assembly arranged within the tube. The external space assembly comprises: a magnetically conductive circumferential pole piece surrounding the tube; a plurality of magnetically conductive supports arranged so as to be distributed around the central axis and extending from the circumferential pole piece up to an outer wall surface of the tube; and a plurality of coils. The vacuum space assembly comprises a plurality of magnetically conductive pole pieces arranged so as to be distributed around the central axis and extending from the tube in the direction of the central axis.



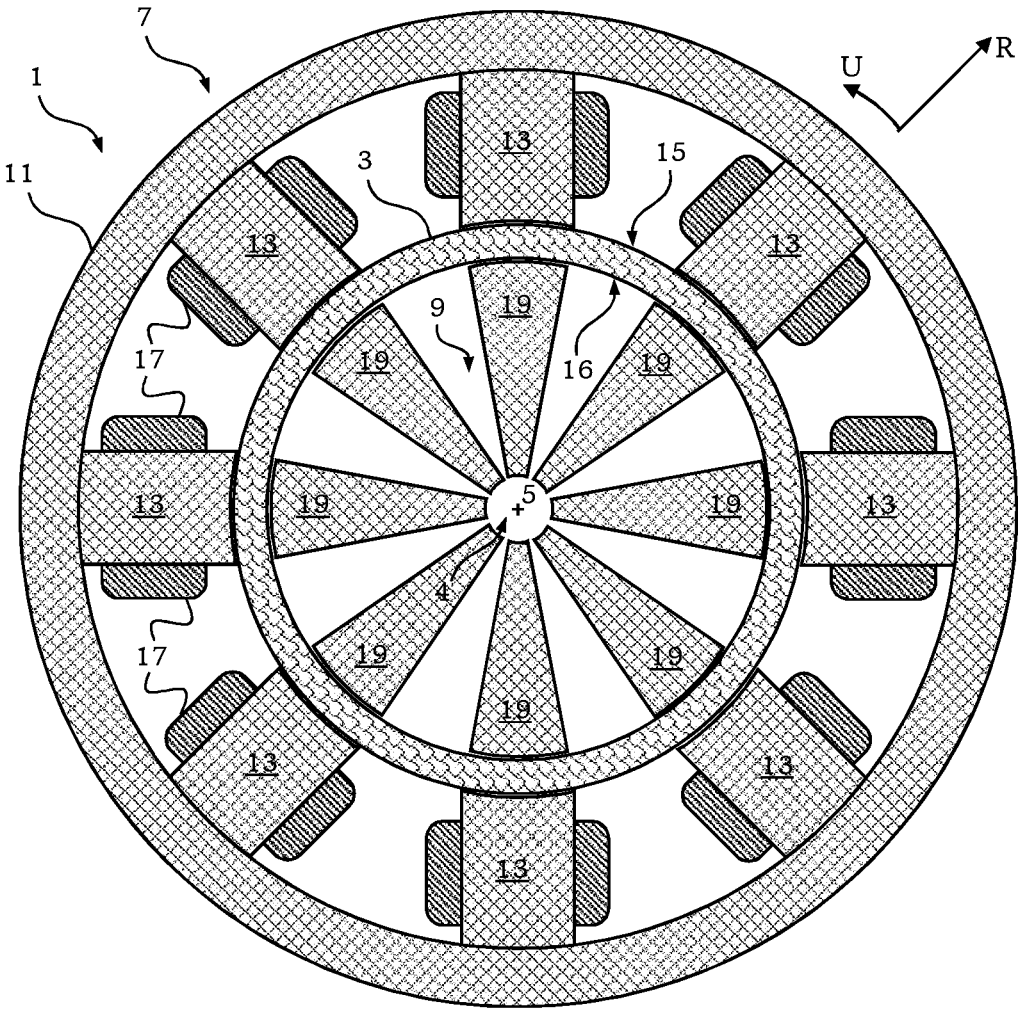


Fig. 1

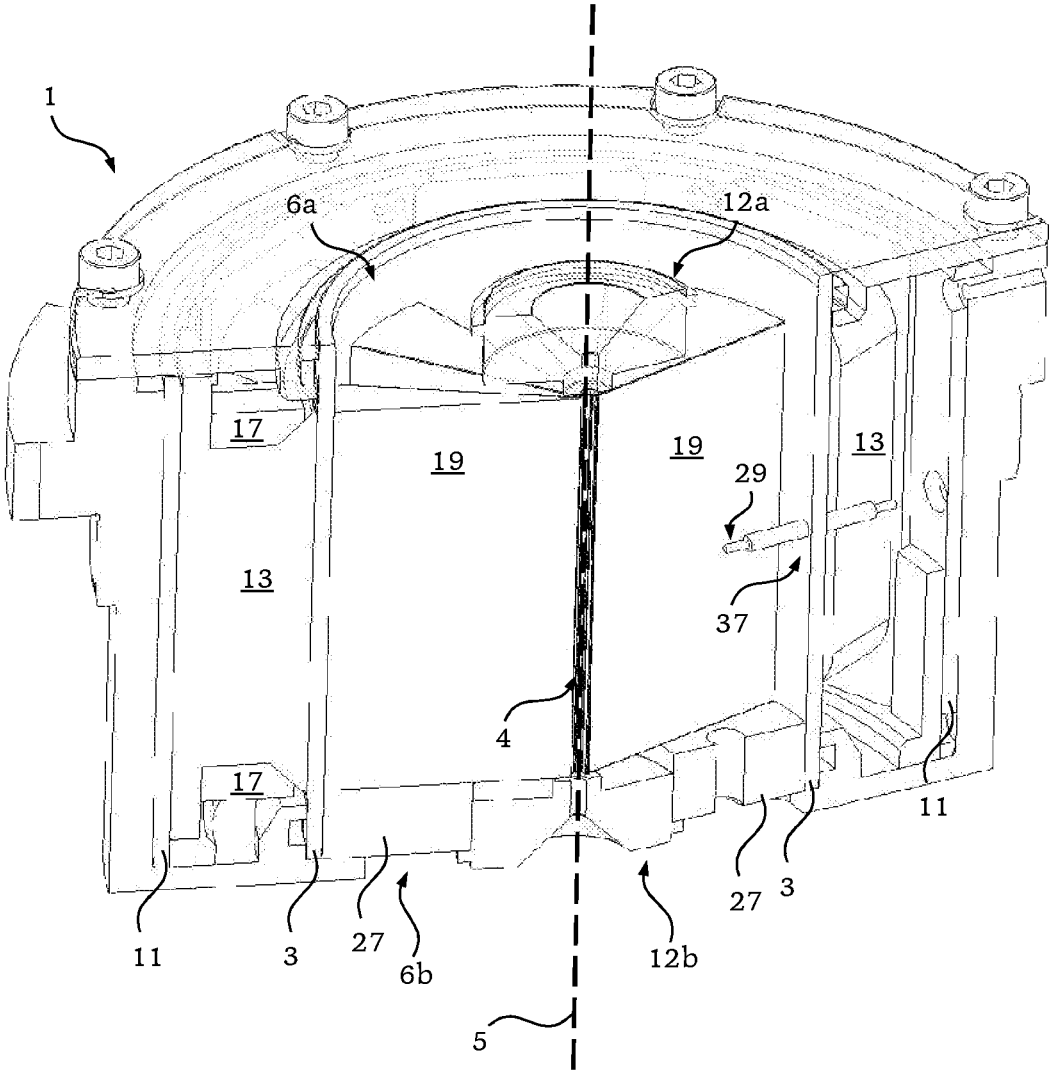


Fig. 2

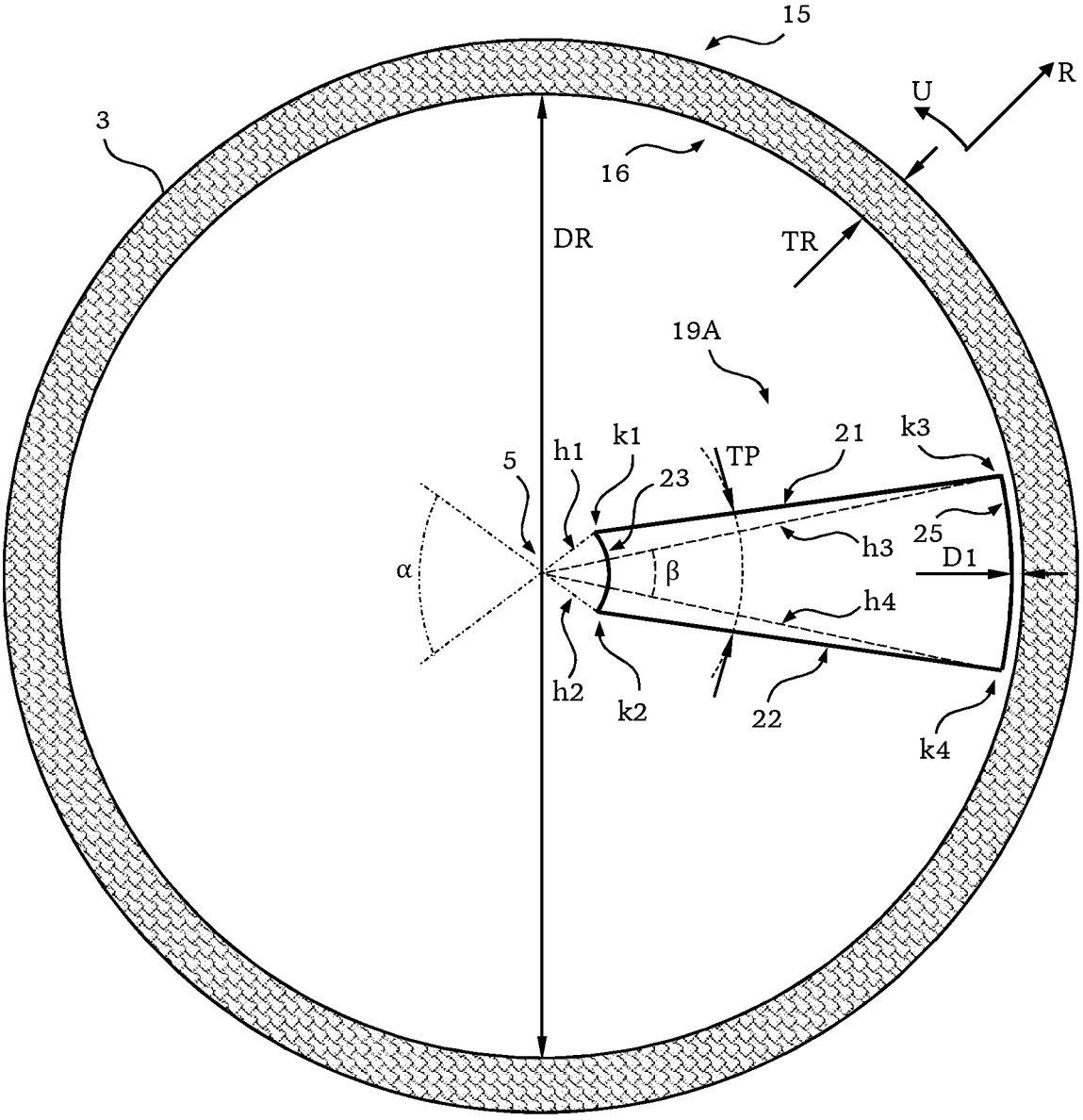


Fig. 3

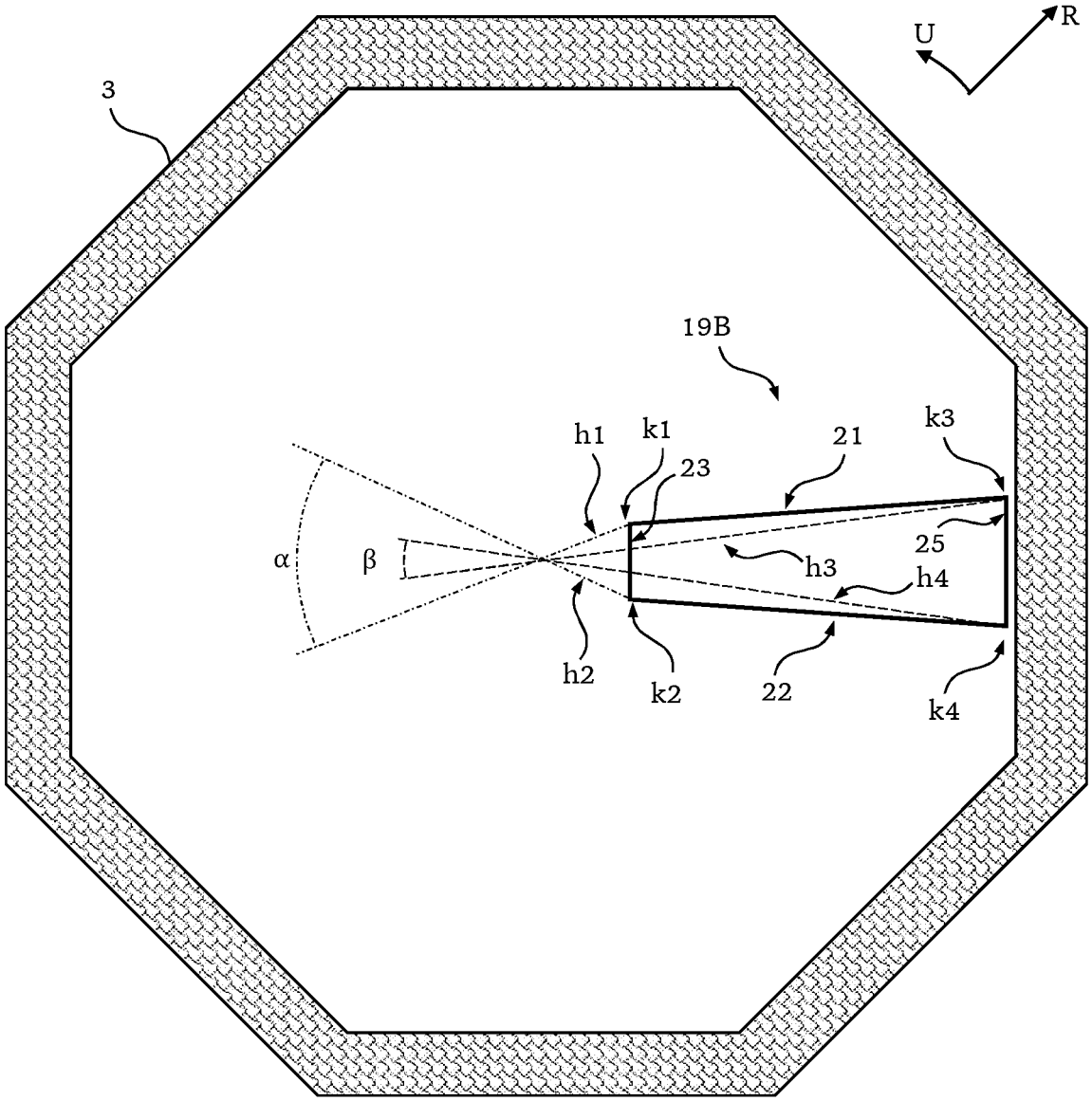


Fig. 4

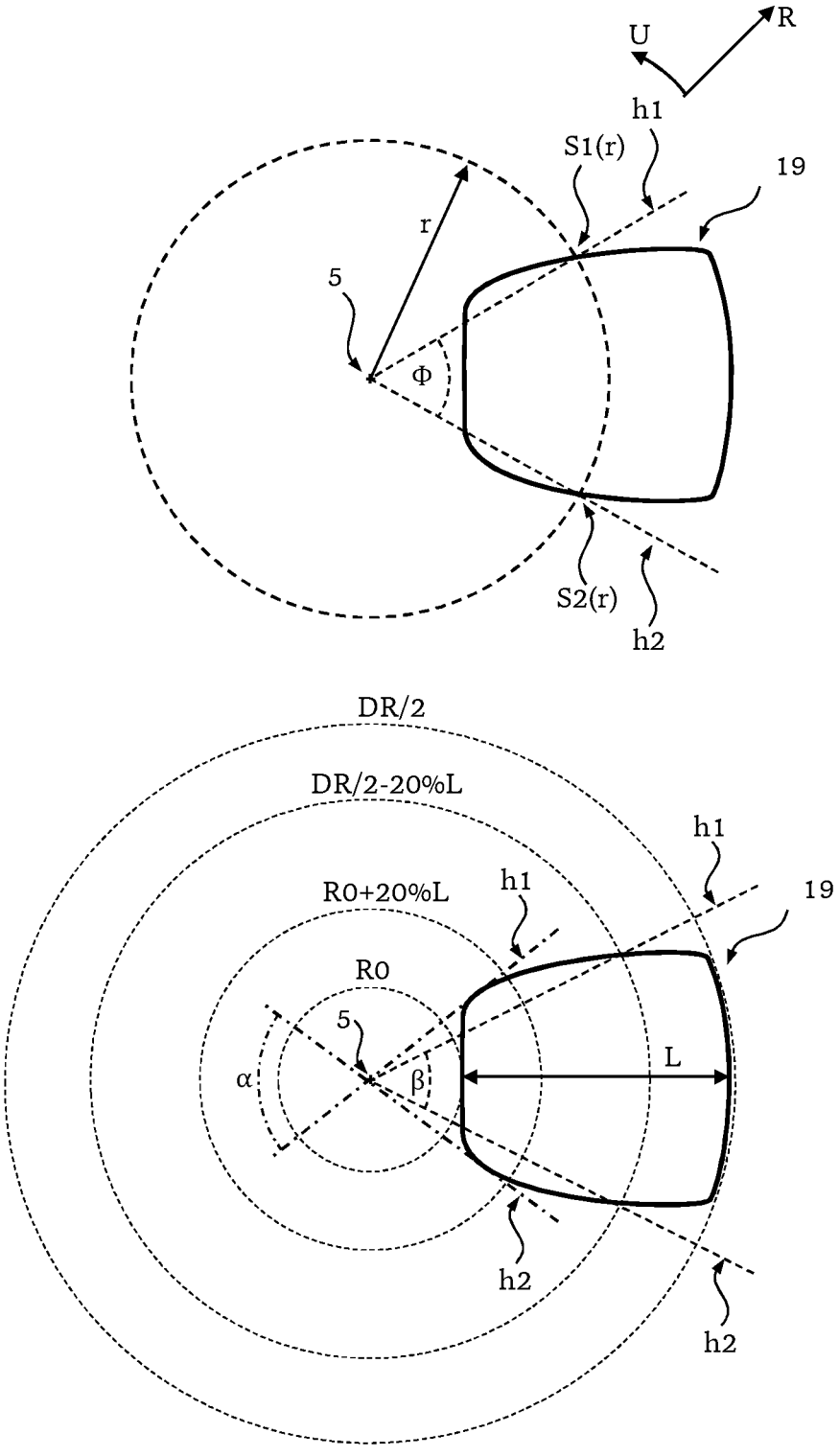


Fig. 5

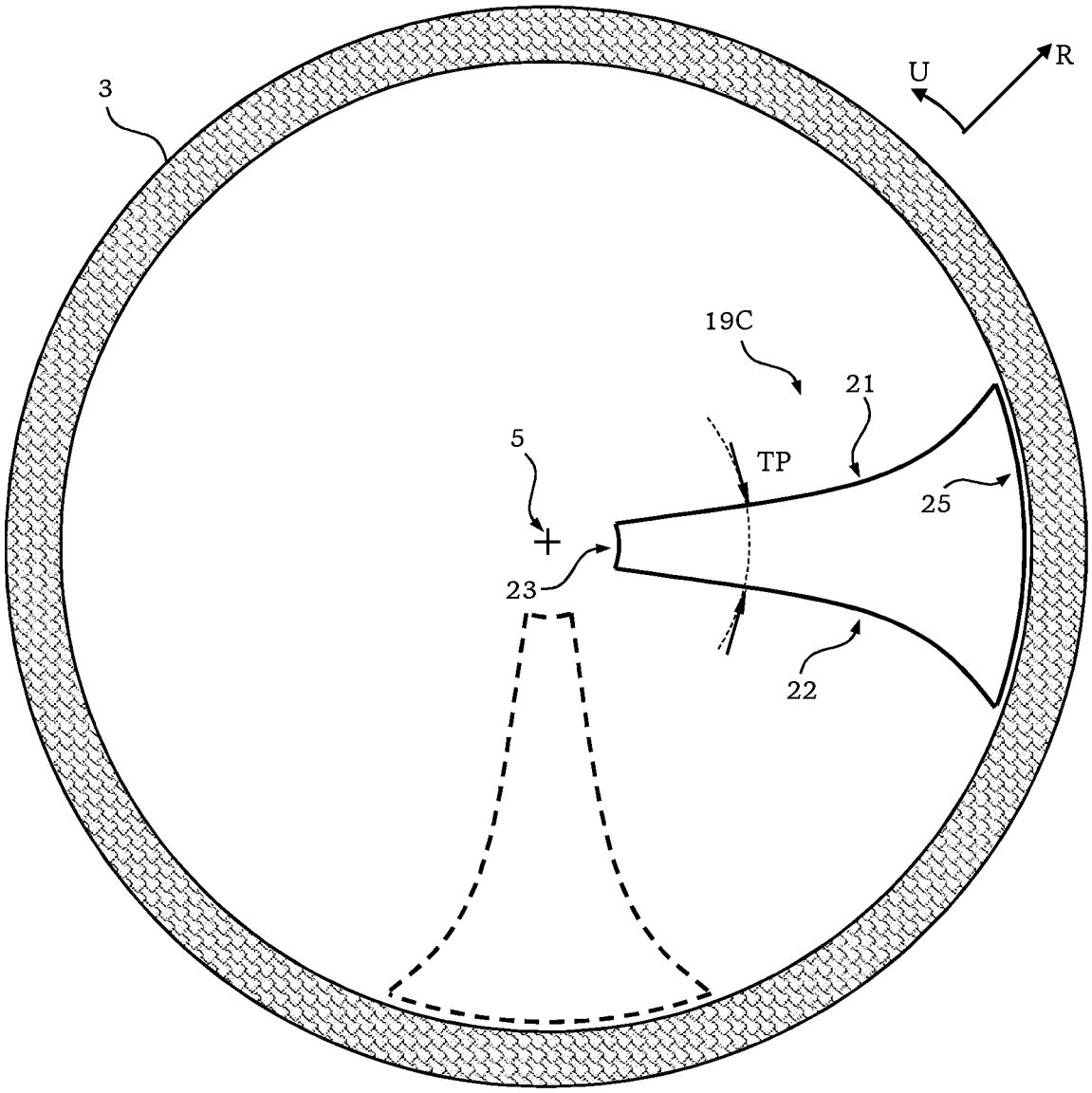


Fig. 6

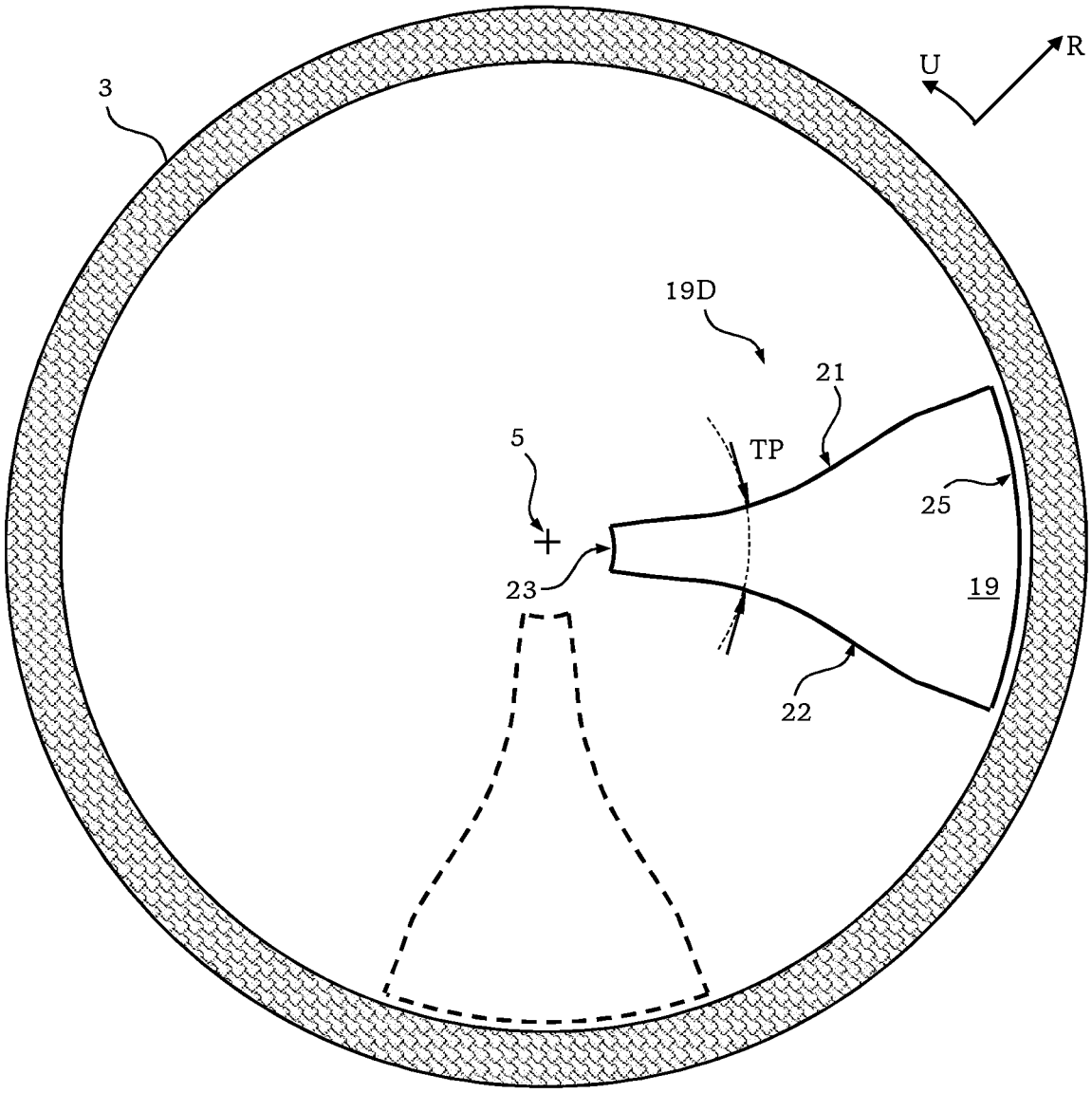


Fig. 7

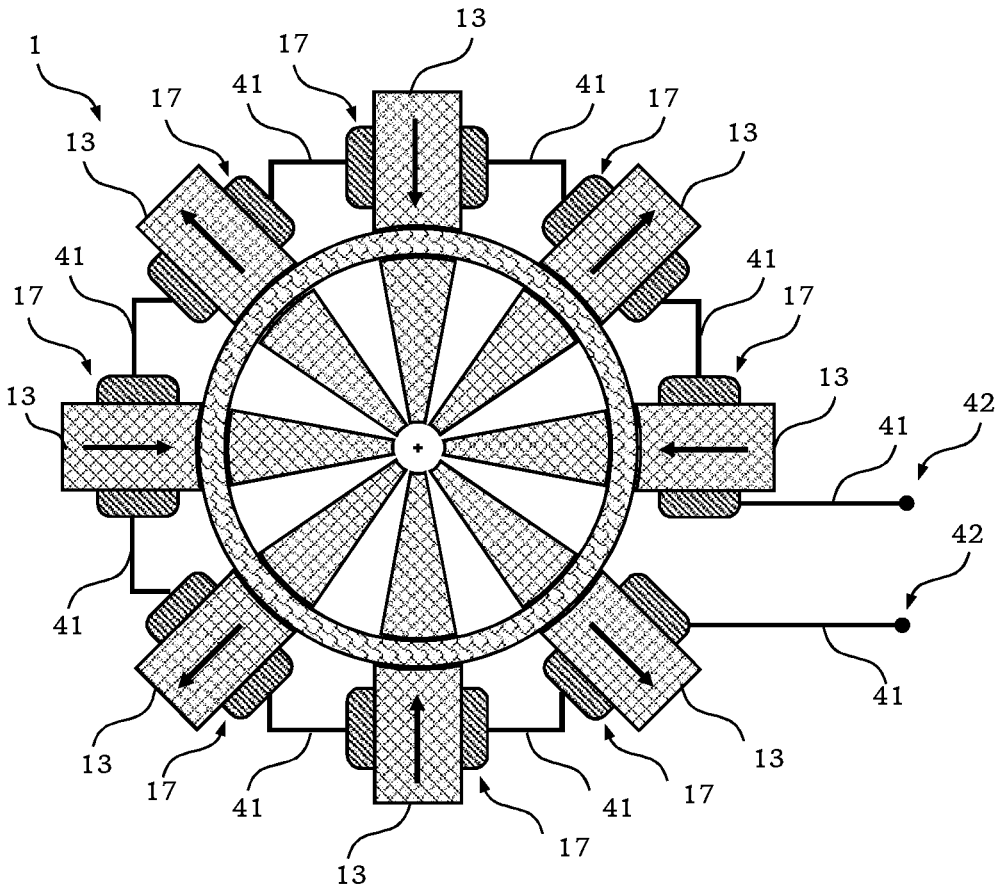


Fig. 8

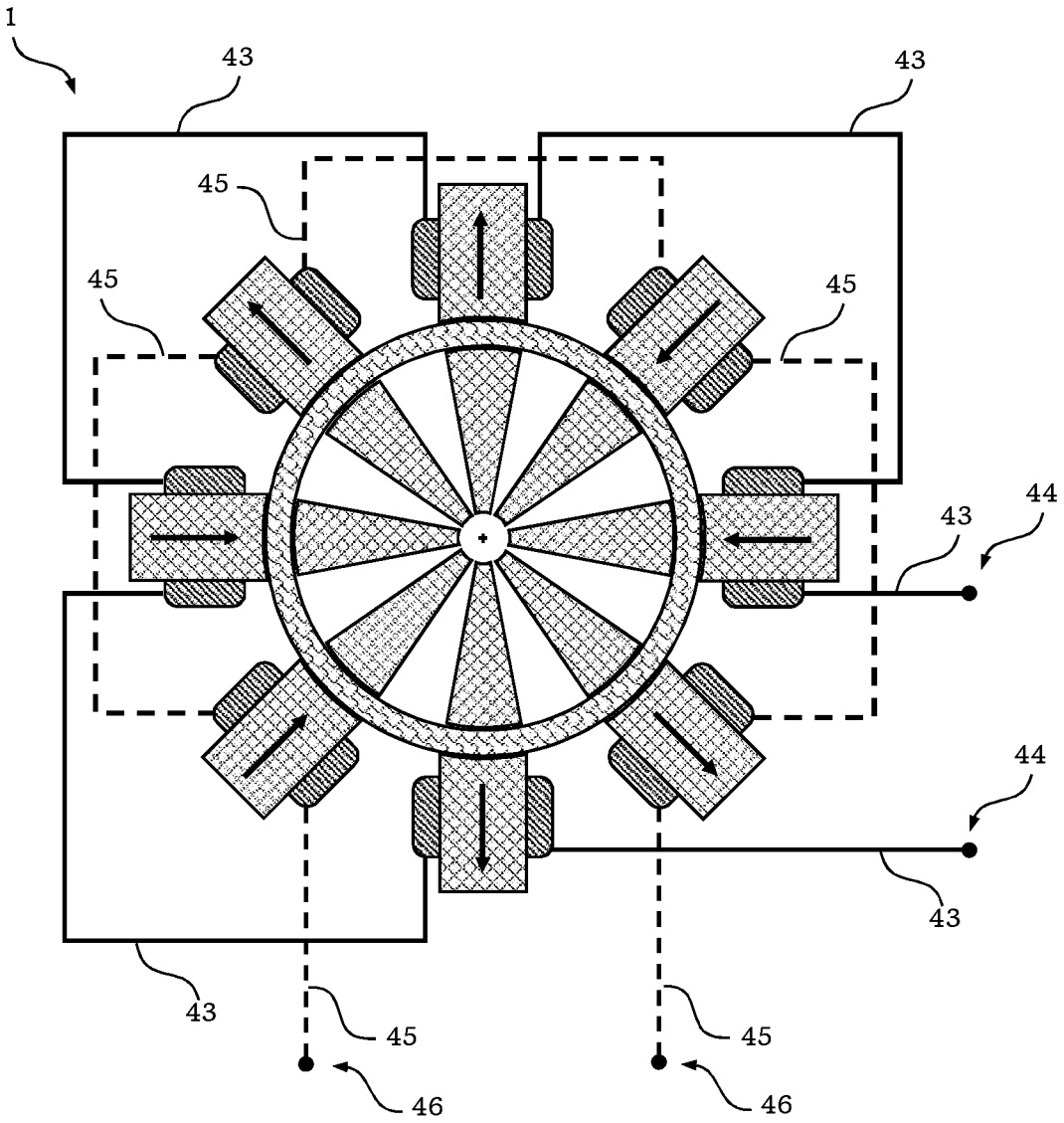


Fig. 9

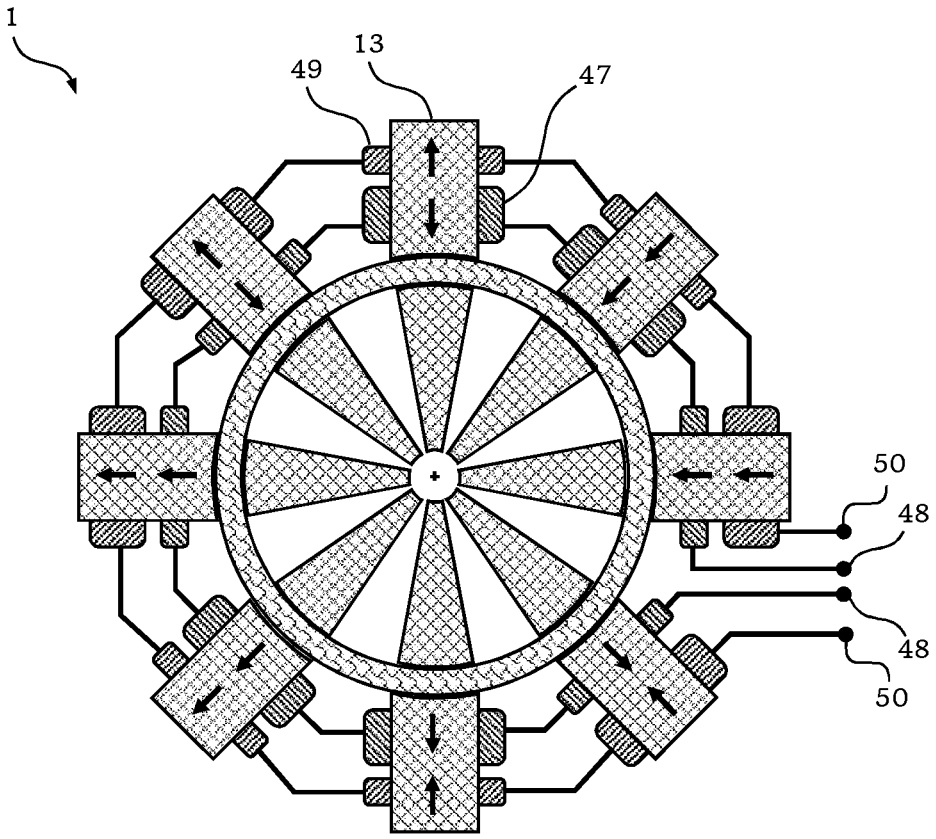


Fig. 10

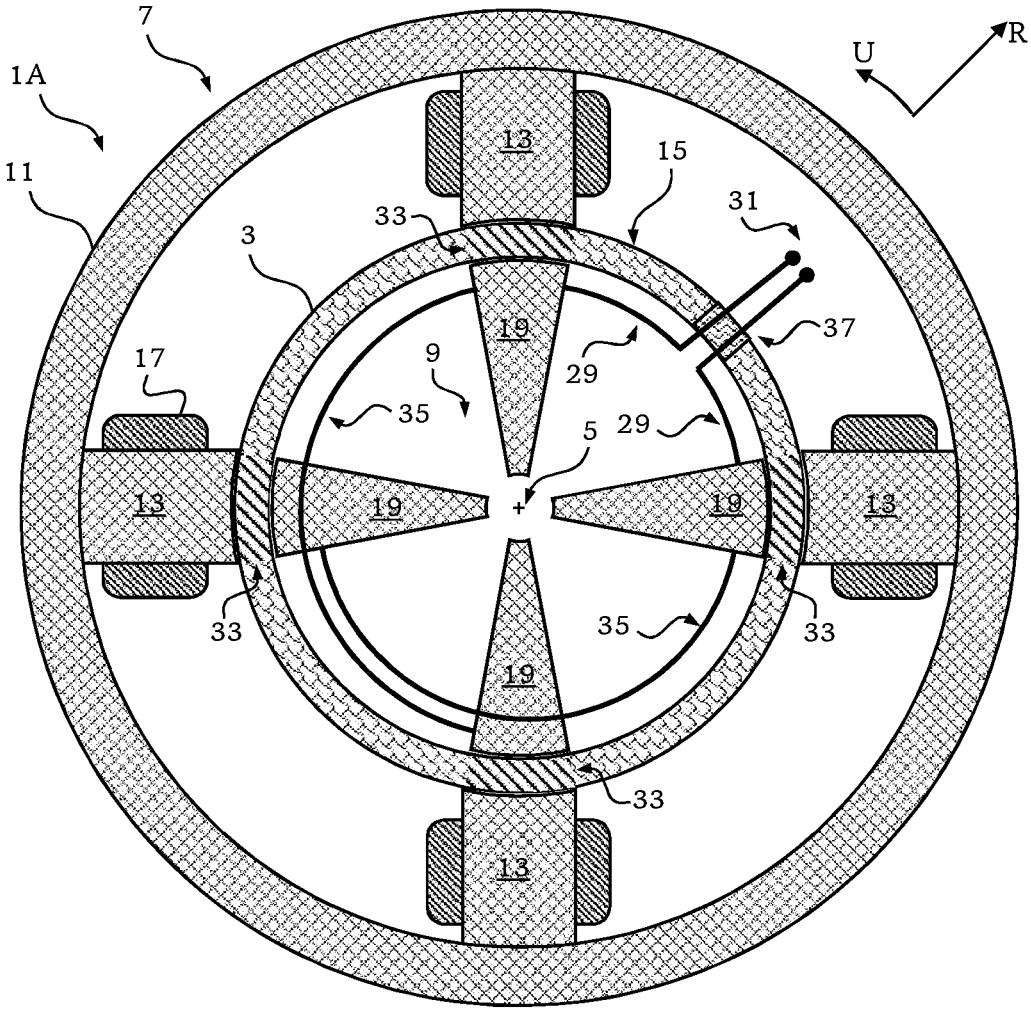

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Fig. 11

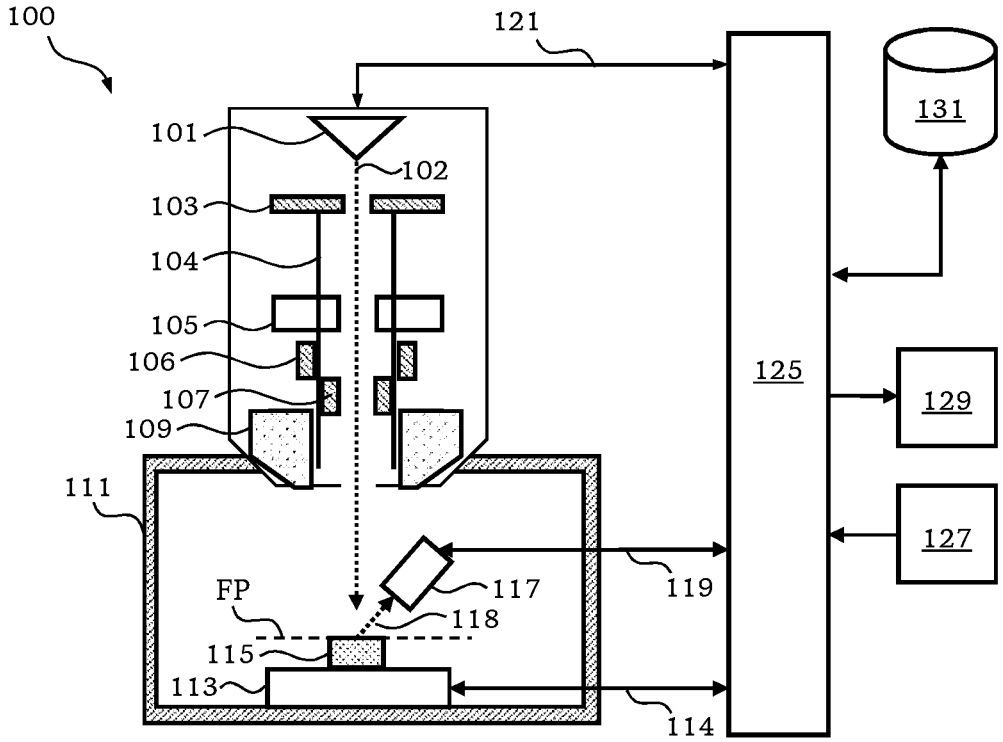

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Fig. 12

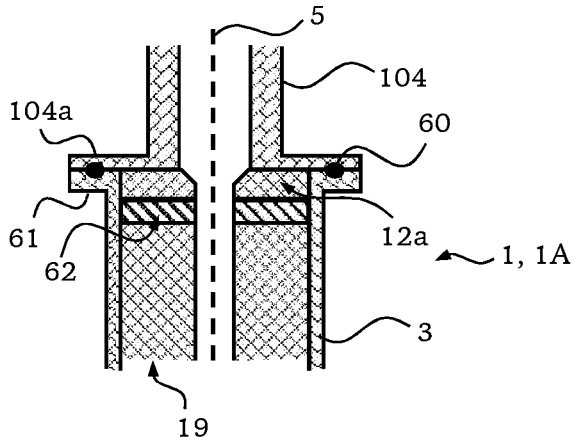

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Fig. 13

MULTIPOLE ELEMENT, IMAGE ERROR CORRECTOR AND PARTICLE BEAM SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims benefit under 35 U.S.C. § 119 to German Application No. 10 2023 111 471.9, filed May 3, 2023. The entire disclosure of this application is incorporated by reference herein.

FIELD

[0002] The present disclosure relates to a multipole element, an image error corrector having the multipole element and a particle beam system having the multipole element. The disclosure also relates to a multipole element for creating a magnetic multipole field or for creating an electric-magnetic multipole field for a particle beam system such as a scanning electron microscope, for example.

BACKGROUND

[0003] Particle beam systems denote systems in which a beam of electrically charged particles (for example electrons or ions) is created and directed at a sample in order to analyse the sample or process the sample. To this end, the particle beam can be raster-scanned over the sample. In the case of an electron beam microscope, interaction products (such as backscattered electrons, secondary electrons, electromagnetic radiation, light, etc.) created by the interaction of the electron beam with the sample are detected in order to analyse the sample. In the case of an ion beam processing device, the ion beam is directed at the sample in order to process the sample, for example remove material from the sample or deposit material on the sample.

[0004] In such particle beam systems, multipole elements are used to manipulate or deflect the particle beam. Examples of conventional multipole elements include stigmators and aberration correctors.

[0005] In multipole elements for creating a magnetic multipole field, the magnetic multipole field is created by coils. The coils are formed by a multiplicity of turns of an insulated wire line. The insulation of the wire line outgasses during operation. The particle beam is guided in a vacuum region and the outgassing of the insulation of the wire line in the vacuum region impairs a vacuum created in the vacuum region during the operation of the multipole element. Encapsulations for the coils in the vacuum region are complicated and expensive. It can therefore be desirable to arrange the coils outside of the vacuum region.

[0006] However, if coils are arranged outside of the vacuum region, magnetic fluxes created by the coils are guided into the vacuum region so that a magnetic multipole field created by the magnetic fluxes can act on the particle beam. EP 0 379 690 A1 discloses a multipole element in which pole pieces pass through a separating wall of the vacuum region and are sealed off by vacuum seals. However, with this configuration, sealing the pole pieces can be complicated and the pole pieces may be provided with an electrical (and hence also magnetic) insulation elsewhere if the magnetic multipole element should be enhanced to form an electric-magnetic multipole element.

SUMMARY

[0007] In an aspect, the disclosure provides a multipole element comprising: a tube surrounding a central axis of the multipole element; an external space assembly arranged outside of the tube in relation to a radial direction emanating perpendicularly from the central axis; and a vacuum space assembly arranged within the tube in relation to the radial direction. The external space assembly comprises: a magnetically conductive circumferential pole piece surrounding the tube in a circumferential direction, the circumferential direction being oriented perpendicularly to the central axis and to the radial direction; a plurality of magnetically conductive supports arranged so as to be distributed around the central axis and extending from the circumferential pole piece against the radial direction up to an outer wall surface of the tube; and a plurality of coils. The vacuum space assembly comprises: a plurality of magnetically conductive pole pieces arranged so as to be distributed around the central axis and extending from the tube against the radial direction.

[0008] The coils of the multipole element are arranged in an external space which is situated outside of the tube in the radial direction. During operation, a vacuum within which for example a particle beam propagates is created in a vacuum region situated within the tube in the radial direction. Since the coils are arranged in the external space, in general, an outgassing by the coils does not influence the vacuum in the vacuum region.

[0009] Magnetic fluxes which flow through the tube and further through the pole pieces are excited in the circumferential pole piece and the supports by the coils. The magnetic flux emerges from the pole pieces at gaps between the pole pieces and forms a magnetic multipole field in the process. Neither the pole pieces nor the supports pass through the tube in the radial direction. Thus, there is no need to use vacuum seals in order to guide the magnetic fluxes excited in the external space into the vacuum region for the purpose of creating the magnetic multipole field. If the tube moreover is electrically nonconductive, then the pole pieces of the vacuum assembly can easily be placed at different, high electric potentials.

[0010] The multipole element can be used in diverse particle beam devices. For example, the multipole element can be used in an electron beam device such as an electron beam microscope or in an ion beam device such as an ion beam processing device.

[0011] Further aspects of the disclosure are an image error corrector comprising at least one multipole element of the type described herein, and a particle beam system comprising at least one image error corrector of the type described herein.

BRIEF DESCRIPTION OF THE FIGURES

[0012] Embodiments of the disclosure are explained in detail below with reference to figures. In detail:

[0013] FIG. **1** shows a schematic cross-sectional view of a multipole element;

[0014] FIG. **2** shows a perspective sectional view of the multipole element;

[0015] FIG. **3** shows a schematic cross-sectional view of a first example of a pole piece of the multipole element;

[0016] FIG. **4** shows a schematic cross-sectional view of a second example of a pole piece of the multipole element;

[0017] FIG. **5** shows a schematic illustration serving to explain the shape of pole pieces;
[0018] FIG. **6** shows a schematic cross-sectional view of a third example of a pole piece of the multipole element;
[0019] FIG. **7** shows a schematic cross-sectional view of a fourth example of a pole piece of the multipole element;
[0020] FIG. **8** shows a schematic illustration of an octupole winding of the multipole element;
[0021] FIG. **9** shows a schematic illustration of quadrupole windings of the multipole element;
[0022] FIG. **10** shows a schematic illustration of dipole windings of the multipole element;
[0023] FIG. **11** shows a schematic cross-sectional view of a further multipole element;
[0024] FIG. **12** shows a schematic illustration of a particle beam system; and
[0025] FIG. **13** shows a schematic cross-sectional view at a connecting point between a beam tube and a multipole element.

DETAILED DESCRIPTION

[0026] A multipole element according to an embodiment of the disclosure is described below with reference to FIGS. **1** and **2**. The multipole element **1** is a magnetic multipole element. This means that the multipole element is configured to create a magnetic multipole field. In the process, equal numbers of magnetic north poles and magnetic south poles are created.
[0027] The geometry of the multipole element **1** is described via a cylindrical coordinate system defined by a longitudinal direction, a radial direction R and a circumferential direction U. To simplify the description, the assumption is made that the longitudinal direction coincides with a central axis **5** of the multipole element **1**. The radial direction R is perpendicular to the longitudinal direction (central axis **5**). The radial direction R emanates perpendicularly from the central axis **5** and extends away from the central axis **5**. The circumferential direction U is always perpendicular to the longitudinal direction and to the radial direction R and parameterized by an angle of rotation about the longitudinal direction.
[0028] FIG. **1** shows a schematic cross-sectional view of the multipole element **1** in a cross-sectional plane oriented perpendicular to the central axis **5**. In FIG. **1**, the central axis **5** and the longitudinal direction are represented by a cross at the centre of the multipole element **1**. The central axis **5** and the longitudinal direction extend perpendicular to the plane of the drawing in FIG. **1**. FIG. **2** shows a perspective sectional view of the multipole element **1**, the section being in a plane spanned by the central axis **5** and the radial direction R.
[0029] The multipole element **1** comprises a tube **3** surrounding the central axis **5** of the multipole element **1**. The multipole element **1** is substantially symmetrical about the central axis **5**. The tube **3** has an upper opening **6***a* at a longitudinally upper end of the tube **3** and has a lower opening **6***b* at a longitudinally lower end of the tube **3**. The tube **3** consists of a wall which circumferentially encloses the central axis **5** at a distance from the central axis **5** and which is longitudinally elongate. A spatial region located within the tube **3** and completely enclosed by the upper opening **6***a*, the lower opening **6***b* and a radially inner wall surface **16** of the tube **3** is referred to as vacuum space. A vacuum is created in the vacuum space during operation. A spatial region located outside of the tube **3** and delimited by the upper opening **6***a*, the lower opening **6***b* and a radially outer wall surface **15** of the tube **3** is referred to as external space. Ambient pressure (no vacuum) is prevalent in the external space during operation.
[0030] The multipole element **1** also comprises an external space assembly **7** arranged outside of the tube **3** in relation to the radial direction R and a vacuum space assembly **9** arranged within the tube **3** in relation to the radial direction R. Accordingly, the external space assembly **7** is situated in the external space, and the vacuum space assembly **9** is situated in the vacuum space. This means that all elements of the external space assembly **7** are arranged outside of the tube **3** in relation to the radial direction R and that all elements of the vacuum space assembly **9** are arranged within the tube **3** in relation to the radial direction R. The tube **3** provides a vacuum separation in the radial direction R between the vacuum space assembly **9** and the external space assembly **7**. This means that a vacuum created vis-à-vis the external space in the vacuum space is maintained by the tube **3** in relation to the radial direction R. When the multipole element **1** is used in a particle beam system, the upper end of the tube **3** and the lower end of the tube **3** are connected to other components of the particle beam system, whereby the vacuum vis-à-vis the external space is also maintained in relation to the longitudinal direction.
[0031] The external space assembly **7** comprises a circumferential pole piece **11** which surrounds the tube **3** in the circumferential direction U. This means that the circumferential pole piece **11** extends in the circumferential direction U around the tube **3** at a distance from the tube **3**.
[0032] The external space assembly **7** also comprises a plurality of supports **13** arranged so as to be distributed around the central axis **5** and extending from the circumferential pole piece **11** against the radial direction R up to an outer wall surface **15** of the tube **3**. This means that the supports **13** extend from the circumferential pole piece **11** to the central axis **5**. The supports **13** extend up to the outer wall surface **15** of the tube **3**. In this case, the supports **13** can penetrate into the tube **3** without piercing fully through the tube **3**. This means that the supports **13** do not pass through the tube **3** in the radial direction R.
[0033] Each support **13** has an inner surface. The inner surface of the support **13** faces the tube **3**. This means that the inner surface of the support **13** is opposite the tube **3** in the radial direction. Each support **13** has an outer surface. The outer surface of the support **13** faces the circumferential pole piece **11**. This means that the outer surface of the support **13** is opposite the circumferential pole piece **11** in the radial direction. Alternatively, the circumferential pole piece **11** and the supports **13** are manufactured from one piece.
[0034] The external space assembly **7** also comprises a plurality of coils **17**. The coils can be carried by the circumferential pole piece **11** and/or by the supports **13**. Each support **13** is assigned one of the coils **17**.
[0035] The vacuum space assembly **9** comprises pole pieces **19** which are arranged so as to be distributed around the central axis **5** and extend from the tube **3** against the radial direction R. This means that the pole pieces **19** extend to the central axis **5** starting from the inner wall surface **16** of the tube **3**. The pole pieces **19** extend up to the inner wall surface **16** of the tube **3**. In this case, the pole pieces **19** can penetrate into the tube **3** without piercing fully through the

tube **3**. The pole pieces **19** are not in contact with the supports **13**. This means that the pole pieces **19** do not pass through the tube **3** in the radial direction R.

[0036] Each pole piece **19** has an inner surface **23** (see FIG. **3**). The inner surface **23** faces the central axis **5**. Each pole piece **19** has an outer surface **25**. The outer surface **25** faces the tube **3**. This means that the outer surface **25** is opposite the tube **3** in the radial direction.

[0037] Each pole piece **19** has an upper surface. The upper surface faces the upper opening **6***a* in the tube **3**. Each pole piece **19** has a lower surface. The lower surface faces the lower opening **6***b* in the tube **3**. Each pole piece **19** has a first lateral surface **21** and a second lateral surface **22**. The first lateral surface **21** faces a first different adjacent pole piece in the circumferential direction U. The second lateral surface **22** faces a second different adjacent pole piece against the circumferential direction U. In the example shown in FIG. **1**, the inner surface **23** and the outer surface **25** are circular segments centred on the central axis **5**; and the lateral surfaces **21**, **22** of the pole pieces **19** are plane faces. In the longitudinal direction, the pole pieces **19** are arranged between an upper termination pole piece **12***a* and a lower termination pole piece **12***b*. The termination pole pieces **12***a*, **12***b* have substantially the shape of thick-walled tubes and are magnetically conductive. Outside the multipole element, this causes the magnetic multipole field to decay quickly in the longitudinal direction.

[0038] A propagation region **4** extends along the central axis **5** over the entire length of the multipole element **1**, extends over the whole circumference of the multipole element **1** in the circumferential direction U and extends from the central axis **5** to the pole pieces **19** in the radial direction R. The propagation region **4** is an empty space in which a particle beam can propagate through the multipole element **1**. The magnetic multipole field is created in the propagation region **4** and thus acts on the particle beam in the propagation region **4**. In particular, the propagation region **4** is delimited by the upper opening **6***a* in the tube **3** and the lower opening **6***b* in the tube **3**.

[0039] The tube **3** is magnetically nonconductive. This element therefore has a high magnetic resistance. For example, the tube **3** is formed from a ceramic or a plastic. Aluminium oxide (Al2O3) is an example of a suitable ceramic. Polyether ether ketone (PEEK) is an example of a suitable plastic.

[0040] The circumferential pole piece **11**, the supports **13** and the pole pieces **19** are each magnetically conductive. These elements therefore have a low magnetic resistance. As a result, magnetic fluxes created by the coils **17** can pass through these elements with little loss, and hence a magnetic multipole field can be created efficiently in the propagation region **4**.

[0041] A magnetic multipole field created by the coils **17** can be created efficiently in the propagation region **4** by way of the above-described configuration of the multipole element **1**. A magnetic flux excited by the coils **17** passes from the coils **17** through the circumferential pole piece **11**, through the supports **13**, through the tube **3**, through the pole pieces **19** and via a gap between the pole pieces **19**. Moreover, the coils **17** are arranged in the external space which is under normal pressure during operation. An outgassing of wire insulations of the coils **17** therefore does not impair a vacuum created in the vacuum space. Moreover, the pole pieces **19** and the supports **13** can be electrically insulated from one another.

[0042] The tube **3** can be a one-piece body. The tube **3** can consist of exactly one material. An internal diameter DR of the tube **3** measured in the radial direction R (see FIG. **3**) can have a value in the range from 20 mm to 200 mm. A length of the tube **3** measured along the central axis **5** can have a value in the range from 20 mm to 200 mm. A wall thickness TR of the tube **3** measured in the radial direction R (see FIG. **3**) can be greater than zero. Within the confines of acceptable stability, the wall thickness TR can be as thin as possible in order to have a magnetic resistance that is as low as possible. The wall thickness TR can be less than 20% of the internal diameter DR of the tube **3**. The wall thickness TR can be less than 10% of the internal diameter DR of the tube **3**. The wall thickness TR can be less than 1% of the internal diameter DR of the tube **3**. In addition or in an alternative, the wall thickness TR cam be less than 5 mm. The wall thickness TR can be less than 2 mm. The wall thickness TR can be less than 1 mm.

[0043] The tube **3** can have a symmetrical cross section in a sectional plane oriented perpendicular to the central axis **5**. In particular, the cross section can have a shape of a circular ring with a constant internal diameter DR and a constant external diameter (see FIG. **3**). In an alternative, the cross section can have a shape of an elliptic ring with an internal diameter DR that varies in the circumferential direction. In a further alternative, the cross section can have a shape of a (regular) polygon or the like. An example of a tube **3** with a cross section in the shape of an octagon is depicted in FIG. **4**. In the longitudinal direction, the cross section of the tube **3** can be constant over the entire length of the tube **3**.

[0044] A diameter of the propagation region **4** measured in the radial direction R can have a value in the range from 1 mm to 20 mm. For example, the diameter of the propagation region **4** is determined as the distance between inner surfaces **23** of two pole pieces **19** that are opposite one another across the central axis **5**. A length of the propagation region **4** in the longitudinal direction corresponds to the length of the tube **3**.

[0045] The circumferential pole piece **11** can be formed in one piece or consist of a plurality of pieces. The circumferential pole piece **11** provides a magnetic guide between supports **13** that are adjacent to one another in the circumferential direction U. To this end, the circumferential pole piece **11** for example is in planar contact with the supports **13**.

[0046] The supports **13** and the circumferential pole piece **11** can be formed in one piece. In this case, the supports **13** are protrusions which project from the circumferential pole piece **11** against the radial direction R. In an alternative, the supports **13** and the circumferential pole piece **11** can be formed as separate pieces.

[0047] The supports **13** can rest against the outer wall surface **15** of the tube **3**. In this case, a distance measured in the radial direction R between the supports **13** and the tube **3** is zero. In an alternative, the supports **13** can have a distance measured in the radial direction R of greater than zero from the outer wall surface **15** of the tube **3**. The distance can be less than 20% of the internal diameter DR of the tube **3** measured in the radial direction R. The distance can be less than 10% of the internal diameter DR of the tube **3**. The distance can be less than 1% of the internal diameter

DR of the tube **3**. In addition or in an alternative, the distance can be less than 5 mm. The distance can be less than 1 mm.

[0048] The distance can be less than 0.1 mm.

[0049] In particular, the supports **13** are arranged with a symmetrical distribution in the circumferential direction U. This means that a difference between angles of rotation at which any two supports **13** that are adjacent in the circumferential direction U are arranged is 360°/(number of supports).

[0050] The pole pieces **19** can rest against the inner wall surface **16** of the tube **3**. In this case, a distance D**1** (see FIG. **3**) measured in the radial direction R between the pole pieces **19** and the tube **3** is zero. In an alternative, the pole pieces **19** can have a distance D**1** measured in the radial direction R of greater than zero from the inner wall surface **16** of the tube **3**. The distance D**1** can be less than 20% of the internal diameter DR of the tube **3** measured in the radial direction R. The distance D**1** can be less than 10% of the internal diameter DR of the tube **3**. The distance D**1** can be less than 1% of the internal diameter DR of the tube **3**. In addition or in an alternative, the distance D**1** can be less than 5 mm. The distance D**1** can be less than 1 mm. The distance D**1** more can be less than 0.1 mm.

[0051] In particular, the pole pieces **19** are arranged with a symmetrical distribution in the circumferential direction U. As a result, the created multipole field has a high degree of symmetry. As shown in FIG. **1**, the pole pieces **19** can lie opposite one another across the central axis **5** in pairs.

[0052] In particular, the pole pieces **19** do not have recesses for mounting the pole pieces **19**. In particular, the pole pieces **19** do not have holes and female threads. As a result, the magnetic resistance of the pole pieces **19** is lower and the magnetic flux through the pole pieces **19** remains undisturbed.

[0053] The number of pole pieces **19** and the number of supports **13** are generally the same but may deviate from one another in special cases. However, the number of pole pieces **19** specifies the maximum number of poles of the magnetic multipole field.

[0054] The supports **13** are arranged relative to the pole pieces **19** such that a magnetic resistance from the supports **13** through the tube **3** into the pole pieces **19** is as small as possible. For example, the supports **13** form an extension of the pole pieces **19** in the radial direction R. For example, the supports **13** are opposite the pole pieces **19** in the radial direction R.

[0055] In the example shown in FIG. **1**, the multipole element **1** is a magnetic octupole. This means that the multipole element **1** is configured to create a magnetic octupole field. However, the multipole element **1** is not restricted to a magnetic octupole. The multipole element **1** can be designed such that the magnetic multipole field has four poles (quadrupole element) or six poles (hexapole element) or 12 poles or 16 poles. An example of a quadrupole element is described below with reference to FIG. **11**.

[0056] As shown in FIG. **2**, the multipole element **1** may also comprise a mount **27** for retaining the pole pieces **19**. In particular, the mount **27** is part of the vacuum space assembly **9**. For example, the mount **27** comprises an upper retaining part (not depicted in FIG. **2** for improved clarity of the drawing) arranged longitudinally above the pole pieces **19** and a lower retaining part arranged longitudinally below the pole pieces **19**. The upper and lower retaining part can be fastened to the tube **3** in any desired way. In this case, the tube **3** contributes to the mount **27** and consequently is a constituent part of the mount **27**. In contrast to mounts known from the prior art, in the case of which the pole pieces **19** are retained by screws, threads, retaining bolts, holes and the like introduced into the mounts, the mount **27** has no negative influence on the magnetic conductivity of the pole pieces **19**. Consequently, the mount **27** enables an efficient creation of the magnetic multipole field.

[0057] Alternatively, the mount **27** can also be provided entirely by the tube **3**. The mount **27** can be formed by adhesively bonding, soldering or clamping the pole pieces **19** to the tube **3** or by similar processes. The mount **27** is made of a magnetically nonconductive material in order to minimize the magnetic flux from a pole piece **19** to another pole piece **19**. The material of the mount **27** further is electrically conductive or nonconductive, depending on desired properties. For a purely magnetic multipole element, this means that the material of the mount **27** can be electrically conductive in order to place the electric potentials of the pole pieces **19** on a common electric potential. For an electric-magnetic multipole element, it can be electrically nonconductive in order to allow different electric potentials of the pole pieces **19**.

[0058] What needs to be taken into account when designing the shape of the pole pieces **19** is that high electrical currents in the coils **17** may lead to an uncontrollably changing magnetic multipole field (so-called drifting) on account of power losses in the coils **17**. A thicker wire reduces the power losses in the coils but increases their volume. However, installation space is limited, and so the coils cannot be arbitrarily large. An efficient transfer of the magnetic fluxes from the supports **13** through the tube **3** to the pole pieces **19** is therefore particularly important. The efficiency can be influenced by the shape of the pole pieces **19**.

[0059] Particular configurations of the pole pieces **19** are described with reference to FIGS. **3** and **4**. FIG. **3** shows a schematic cross-sectional view of a first example of a pole piece **19**A of the multipole element **1** in a cross-sectional plane oriented perpendicular to the central axis **5**. FIG. **4** shows a schematic cross-sectional view of a second example of a pole piece **19**B of the multipole element **1** in a cross-sectional plane oriented perpendicular to the central axis **5**. To simplify the explanations, FIGS. **3** and **4** each show only a single pole piece **19**A and **19**B, respectively, even though the multipole element **1** always comprises a plurality of pole pieces.

[0060] FIG. **3** shows the pole piece **19**A with its inner surface **23**, its outer surface **25**, its first lateral surface **21** and its second lateral surface **22**. At a first edge k**1**, the first lateral surface **21** of the pole piece **19**A abuts the inner surface **23** of the pole piece **19**A. At a second edge k**2**, the second lateral surface **22** of the pole piece **19**A abuts the inner surface **23** of the pole piece **19**A. A first half-line h**1** lies in the cross-sectional plane (plane of the drawing), emanates from the central axis **5** and runs through the first edge k**1**. A second half-line h**2** lies in the cross-sectional plane, emanates from the central axis **5** and runs through the second edge k**2**. An acute angle between the first half-line h**1** and the second half-line h**2** is referred to as inner coverage angle α. To simplify the illustration, the first half-line h**1** and the second half-line h**2** were extended beyond the central axis **5** and the inner coverage angle α was plotted on the extended portions.

[0061] At a third edge k**3**, the first lateral surface **21** of the pole piece **19**A abuts the outer surface **25** of the pole piece **19**A. At a fourth edge k**4**, the second lateral surface **22** of the pole piece **19**A abuts the outer surface **25** of the pole piece **19**A. A third half-line h**3** lies in the cross-sectional plane, emanates from the central axis **5** and runs through the third edge k**3**. A fourth half-line h**4** lies in the cross-sectional plane, emanates from the central axis **5** and runs through the fourth edge k**4**. An acute angle between the third half-line h**3** and the fourth half-line h**4** is referred to as outer coverage angle β.

[0062] The peculiarity of the pole piece **19**A is that the inner coverage angle α and the outer coverage angle β have values that differ from one another. For example, this means that the inner coverage angle α and the outer coverage angle β differ from one another by at least 5°. In particular, the pole pieces **19** are shaped such that the inner coverage angle α is greater than the outer coverage angle β.

[0063] The inner coverage angle α influences the magnetic resistance between the inner surface **23** of the pole pieces **19** and the propagation region **4**: As the inner coverage angle α increases there is a reduction in the magnetic resistance between the inner surface **23** of the pole pieces **19** and the propagation region **4**. The smaller the magnetic resistance, the smaller the excitation power used to create a desired magnetic field strength of the multipole field in the propagation region **4**. In other words: The efficiency of creating the magnetic multipole field is higher the smaller the magnetic resistance is between the inner surface **23** of the pole pieces **19** and the propagation region **4**.

[0064] The outer coverage angle β influences the magnetic resistance between the outer surface **25** of the pole pieces **19** and the supports **13**: As the outer coverage angle β increases there is a reduction in the magnetic resistance between the outer surface **25** of the pole pieces **19** and the supports **13**. The smaller the magnetic resistance, the smaller the excitation power used to create a desired magnetic field strength of the multipole field in the propagation region **4**. In other words: The efficiency of creating the magnetic multipole field is higher the smaller the magnetic resistance is between the outer surface **25** of the pole pieces **19** and the supports **13**.

[0065] The distance between adjacent pole pieces **19** in the circumferential direction U, which is also influenced by the inner coverage angle α and the outer coverage angle β, influences the magnetic resistance between the pole pieces **19**. This magnetic resistance is subordinate to the magnetic resistance present between the pole pieces **19** and the supports **13** and guides the magnetic field around the propagation region **4**. Thus, the following applies: the smaller the magnetic resistance between the pole pieces **19**, the greater the excitation power used to create a desired magnetic field strength of the multipole field in the propagation region **4**. In other words: The efficiency of creating the magnetic multipole field is higher the greater the magnetic resistance is between adjacent pole pieces **19** in the circumferential direction U.

[0066] The inventors have recognized that a skillful choice of the inner coverage angle α and outer coverage angle β leads to a particularly efficient creation of the magnetic multipole field.

[0067] FIG. **4** shows an example of a tube **3** with an octagonal cross section. Matched thereto, the inner surface **23** and the outer surface **25** of the pole piece **19**B are not circular segments but plane faces. In the pole piece **19**B shown in FIG. **4**, too, the inner coverage angle α and the outer coverage angle β have values that differ from one another. The inner coverage angle α is greater than the outer coverage angle β.

[0068] Examples of shapes of pole pieces are characterized on the basis of a sector width below. The sector width is defined as the ratio of 360° to the number of pole pieces **19**. The sector width is 90° in the case of a quadrupole element. The sector width is 60° in the case of a hexapole element. The sector width is 45° in the case of an octupole element.

[0069] For example, the pole pieces **19** are shaped such that a difference between the inner coverage angle α and the outer coverage angle β is at least 10% of the sector width, corresponding to a difference of at least 9° in the case of a quadrupole element and corresponding to a difference of at least 4.5° in the case of an octupole element. The difference between the inner coverage angle α and the outer coverage angle β can be at least 20% of the sector width, corresponding to a difference of at least 18° in the case of a quadrupole element and corresponding to a difference of at least 9° in the case of an octupole element.

[0070] For example, the pole pieces **19** are shaped such that a ratio of the inner coverage angle α to the sector width has a value in the range of 75% to 95%, corresponding to an inner coverage angle α in the range of 67.5° to 85.5° in the case of a quadrupole element and corresponding to an inner coverage angle α in the range of approx. 34° to approx. 43° in the case of an octupole element. The ratio of the inner coverage angle α to the sector width can have a value in the range of 80% to 90%, corresponding to an inner coverage angle α in the range of 72° to 81° in the case of a quadrupole element and corresponding to an inner coverage angle α in the range of 36° to 40.5° in the case of an octupole element.

[0071] For example, the pole pieces **19** are shaped such that a ratio of the outer coverage angle β to the sector width has a value in the range of 35% to 75%, corresponding to an outer coverage angle β in the range of 31.5° to 67.5° in the case of a quadrupole element and corresponding to an outer coverage angle β in the range of approx. 16° to approx. 34° in the case of an octupole element. The ratio of the outer coverage angle β to the sector width can have a value in the range of 50% to 70%, corresponding to an outer coverage angle β in the range of 45° to 63° in the case of a quadrupole element and corresponding to an outer coverage angle β in the range of approx. 22.5° to approx. 31.5° in the case of an octupole element.

[0072] Determining the inner coverage angle α and the outer coverage angle β in accordance with the definition above may be difficult in the case of pole pieces **19** with rounded-off edges k**1** to k**4**. Therefore, an alternative or more general definition for the inner coverage angle α and the outer coverage angle β is explained below with reference to FIG. **5**.

[0073] The upper part in FIG. **5** shows an exemplary pole piece **19** with rounded-off edges, in a cross-sectional plane oriented perpendicularly to the central axis **5**. The central axis **5** is perpendicular to the plane of the drawing in FIG. **5**. A radius-dependent coverage angle Φ(r) specifies an acute angle between a first half-line h**1** and a second half-line h**2**. The first half-line h**1** lies in the cross-sectional plane oriented perpendicularly to the central axis **5**, emanates from the central axis **5** and runs through a first point of intersec-

tion S1(r). The second half-line h2 lies in the cross-sectional plane, emanates from the central axis **5** and runs through a second point of intersection S2(r). The first point of intersection S1(r) and the second point of intersection S2(r) are (radius-dependent) points of intersection of the surface of the pole piece **19** with a circle which lies in the cross-sectional plane, has the radius r and has its centre on the central axis **5**. This means that the coverage angle $\Phi(r)$ has different values depending on the radius r of the (imaginary) circle, the values depending on the specific shape of the pole piece **19**.

[0074] The inner coverage angle α is intended to specify the coverage angle $\Phi(r)$ in a region in the proximity of the inner surface **23** or in the proximity of the propagation region **4**. To this end, the inner coverage angle α is for example defined as follows: The inner coverage angle α is the largest radius-dependent coverage angle $\Phi(r)$ in a range $R0 \leq r < R0+20\%\ L$, where R**0** is the smallest distance measured in the radial direction R between the surface of the pole piece **19** and the central axis **5** and where L is the maximum length of the pole piece **19** measured in the radial direction R. The range $R0 \leq r < R0+20\%\ L$ denotes a range for the radius r with a lower limit at R**0** and an upper limit at R**0**+20% L. An example of this definition is depicted in the lower part of FIG. **5**, wherein the half-lines h**1** and h**2** have been depicted extended beyond the central axis **5** in order to improve the clarity of the figure. The half-lines h**1** and h**2**, which define the inner coverage angle α, are depicted as dash-dotted lines. To illustrate the range $R0 \leq r < R0+20\%\ L$, FIG. **5** depicts a circle with radius R**0** around the central axis **5** and a circle with radius R**0**+20% L around the central axis **5**.

[0075] The outer coverage angle α is intended to specify the coverage angle $\Phi(r)$ in a region in the proximity of the outer surface **25** or in the proximity of the tube **3**. To this end, the outer coverage angle β is for example defined as follows: The outer coverage angle is the largest radius-dependent coverage angle $\Phi(r)$ in a range $DR/2-20\%\ L < r \leq DR/2$, where DR is the internal diameter of the tube **3** and where L is the maximum length of the pole piece **19** measured in the radial direction R. The range $DR/2-20\%\ L < r \leq DR/2$ denotes a range for the radius r with a lower limit at DR/2−20% L and an upper limit at DR/2. An example of this definition is depicted in the lower part of FIG. **5**. The half-lines h**1** and h**2**, which define the outer coverage angle β, are depicted as dashed lines. To illustrate the range $DR/2-20\%\ L < r \leq DR/2$, FIG. **5** depicts a circle with radius DR/2−20% L around the central axis **5** and a circle with radius DR/2 around the central axis **5**.

[0076] The aforementioned values and value ranges for the inner coverage angle α and the outer coverage angle β apply in particular to pole pieces **19** whose lateral surfaces **21**, **22** are plane faces. However, the aforementioned values and value ranges for the inner coverage angle α and the outer coverage angle β may also apply to pole pieces **19** whose lateral surfaces **21**, **22** are curved or bent.

[0077] Further particular configurations of the pole pieces **19** are described with reference to FIGS. **6** and **7**. FIG. **6** shows a schematic cross-sectional view of a third example of a pole piece **19**C of the multipole element **1** in a cross-sectional plane oriented perpendicular to the central axis **5**. FIG. **7** shows a schematic cross-sectional view of a fourth example of a pole piece **19**D of the multipole element **1** in a cross-sectional plane oriented perpendicular to the central axis **5**. To simplify the explanations, FIGS. **6** and **7** each show only a single pole piece **19**C and **19**D, respectively, (and a pole piece adjacent thereto in the circumferential direction U is indicated using dashed lines) even though the multipole element **1** always comprises a plurality of pole pieces.

[0078] The pole piece **19**C is distinguished in that the two lateral surfaces **21**, **22** of the pole piece **19**C are not plane faces. Instead, the two lateral surfaces **21**, **22** of the pole piece **19**C are curved. Rather than having a curved shape, the lateral surfaces **21**, **22** can also be bent.

[0079] On account of this shape of the lateral surfaces **21**, **22** of the pole pieces **19**, a distance over large regions of the lateral surfaces **21**, **22** is greater between pole pieces **19** that are adjacent in the circumferential direction U than in the case of lateral surfaces **21**, **22** with a plane face shape. As a result, the local magnetic resistance of a space between adjacent pole pieces with curved or bent lateral surfaces is greater than the local magnetic resistance of a space between adjacent pole pieces which have lateral surfaces in the form of plane faces. In comparison with pole pieces with lateral surfaces in the form of plane faces, a smaller portion of a magnetic flux guided in the pole piece emerges via the lateral surfaces **21**, **22** between pole pieces with curved or bent lateral surfaces as a result. This increases the portion of the magnetic flux which is guided in the pole piece and emerges via the inner surface **23**, leading to a stronger magnetic multipole field in the case of the same excitation (current through the coils **17**). Hence, the efficiency of creating the magnetic multipole field is increased by the curved or bent side faces **21**, **22** of the pole pieces **19**.

[0080] A measure for this efficiency is the curve of a coverage angle $\Phi(r)$ as a function of radius r. For this purpose, a mid coverage angle γ is for example defined as follows: The mid coverage angle is the smallest radius-dependent coverage angle $\Phi(r)$ in a range $R0+20\%\ L < r < DR/2-20\%\ L$, where R**0** is the smallest distance measured in the radial direction R between the surface of the pole piece **19** and the central axis **5**, where DR is the internal diameter of the tube **3** and where L is the maximum length of the pole piece **19** measured in the radial direction R. The range $R0+20\%\ L < r < DR/2-20\%\ L$ denotes a range for the radius r with a lower limit at R**0**+20% L and an upper limit at DR/2−20% L.

[0081] A significant increase in the efficiency of the creation of the magnetic multipole field by curved or bent side faces arises if the mid coverage angle γ is both smaller than the inner coverage angle α and smaller than the outer coverage angle β. In this context, the outer coverage angle β may also assume values greater than 75% of the sector width. In this case, the inner coverage angle α and outer coverage angle β may have different values or the same value.

[0082] Details regarding the coils **17** of the multipole element **1** are described below with reference to FIGS. **8** to **10**. Each coil **17** may comprise one or more windings. Each winding consists of at least one turn. A turn is a wire loop. Normally each winding consists of a multiplicity of turns. The number of turns per winding is typically several tens to several thousands of turns. Increasing the number of turns reduces a current used to flow through the winding in order to create a certain magnetic flux.

[0083] If a coil **17** comprises a plurality of windings, then the plurality of windings are arranged such that the magnetic

fluxes created by the plurality of windings are superimposed in space. A plurality of windings are electrically insulated from one another and operated by separate current sources. A winding may extend over a plurality of the coils **17**.

[0084] FIG. **8** shows a schematic illustration of an octupole winding **41** of the multipole element **1**. The circumferential pole piece **11** is not depicted here (and not in FIGS. **9** and **10** either) for the sake of clarity. The octupole winding **41** comprises two connectors **42**, at which the octupole winding **41** is connectable to a current source. The octupole winding **41** consists of a single electrical conductor (for example a wire or a bundle of wires). The electrical conductor of the octupole winding **41** extends over all eight coils **17** of the multipole element **1** designed as an octupole element. The number of turns of the octupole winding **41** at each coil **17** (i.e. the number of turns of the octupole winding **41** wound around each of the supports **13** or each portion of the circumferential pole piece **11**) is the same and totals 200, for example. An arrow oriented either in the radial direction R or against the radial direction R is depicted in each of the supports **13**. The orientation of the arrow specifies the winding sense of the turns of the octupole winding **41** at the respective coil **17**. For example, the tip of the arrow specifies the position of the magnetic north pole, and the start of the arrow specifies the magnetic south pole. As depicted by the orientations of the arrows in FIG. **8**, the winding senses of each pair of coils **17** adjacent to one another in the circumferential direction U are opposite to one another. Accordingly, in the circumferential direction U, the arrows are alternately oriented in the radial direction R and against the radial direction R. Using the octupole winding **41**, it is possible to create a magnetic octupole field in the propagation region **4**.

[0085] FIG. **9** shows a schematic illustration of quadrupole windings of the multipole element **1**. A first quadrupole winding **43** is depicted by a solid line. The first quadrupole winding **43** comprises two connectors **44**, at which the first quadrupole winding **43** is connectable to a current source. The first quadrupole winding **43** consists of a single electrical conductor. The electrical conductor of the first quadrupole winding **43** extends over four of the eight coils **17** of the multipole element **1** designed as an octupole element. When counting sequentially in the circumferential direction, the first quadrupole winding **43** extends over all odd coils **17** of the multipole element **1**. The winding senses of each pair of coils **17** over which the first quadrupole winding **43** extends and which are adjacent in the circumferential direction U are opposite one another, as depicted by appropriate arrows in FIG. **9**. The number of turns of the quadrupole winding **43** at each of the coils **17** is the same and totals 1000, for example. Using the first quadrupole winding **43**, it is possible to create a first magnetic quadrupole field in the propagation region **4**.

[0086] A second quadrupole winding **45** is depicted by a dashed line. The second quadrupole winding **45** comprises two connectors **46**, at which the second quadrupole winding **45** is connectable to a current source. The second quadrupole winding **45** consists of a single electrical conductor. The electrical conductor of the second quadrupole winding **45** extends over four of the eight coils **17** of the multipole element **1** designed as an octupole element. When counting sequentially in the circumferential direction, the second quadrupole winding **45** extends over all even coils **17** of the multipole element **1**. The winding senses of each pair of coils **17** over which the second quadrupole winding **45** extends and which are adjacent in the circumferential direction U are opposite one another, as depicted by appropriate arrows in FIG. **9**. Using the second quadrupole winding **45**, it is possible to create a second magnetic quadrupole field in the propagation region **4**. Effectively, the first magnetic quadrupole field and the second magnetic quadrupole field yield a magnetic quadrupole field that is rotatable in the circumferential direction U.

[0087] FIG. **10** shows a schematic illustration of dipole windings of the multipole element **1**. A first dipole winding **47** comprises two connectors **48**; a second dipole winding **49** comprises two connectors **50**. Hence the two dipole windings **47** and **49** are connectable to separate current sources. The dipole windings **47** and **49** each consist of a single electrical conductor.

[0088] The first dipole winding **47** has a multiplicity of turns at each support **13**. The number of turns of the first dipole winding **47** at a respective support **13** is represented schematically by the thickness of the first dipole winding **47** at the respective support **13**. The orientation of the turns of the first dipole winding **47** at a respective support **13** is represented schematically by an arrow (located to the inside in the radial direction R) at the respective support **13**. For example, the turns of the first dipole winding **47** have the following configuration: +100, +100, +41, −41, −100, −100, −41, +41, where the sign specifies the orientation of the turn and the numeral specifies the number of turns at the respective support **13**, starting with the support **13** depicted top centre and proceeding clockwise.

[0089] The second dipole winding **49** has a multiplicity of turns at each support **13**. The number of turns of the second dipole winding **49** at a respective support **13** is represented schematically by the thickness of the second dipole winding **49** at the respective support **13**. The orientation of the turns of the second dipole winding **49** at a respective support **13** is represented schematically by an arrow (located to the outside in the radial direction R) at the respective support **13**. For example, the turns of the second dipole winding **49** have the following configuration: −41, +41, +100, +100, +41, −41, −100, −100, where the sign specifies the orientation of the turn and the numeral specifies the number of turns at the respective support **13**, starting with the support **13** depicted top centre and proceeding clockwise. Using the dipole windings **47** and **49**, it is possible to create a magnetic dipole field, rotatable as desired, in the propagation region **4**.

[0090] Such dipole fields can also be created with different configurations of the number of turns and orientations of the turns. The present embodiment only shows an embodiment for implementing a dipole winding.

[0091] A further multipole element **1**A according to an embodiment of the disclosure is described below with reference to FIG. **11**. In contrast to the multipole element **1** described above, the multipole element **1**A is an electric-magnetic multipole element. This means that the multipole element **1**A is configured to create a magnetic multipole field and an electric multipole field. In this context, the same number of magnetic north poles and magnetic south poles of the magnetic multipole field are created; and the same number of positive poles and negative poles of the electric multipole field are created. Moreover, the number of poles of the magnetic multipole field equals the number of poles of the electric multipole field. In particular, the magnetic mul-

tipole field and the electric multipole field are created such that they are superimposed on one another in space.

[0092] A further difference to the multipole element **1**, which is an octupole element, is that the multipole element **1**A is a quadrupole element having four pole pieces **19** and four supports **13**. Four is the maximum number of electric field poles, and four is the maximum number of magnetic field poles. The multipole element **1**A may have a different number of poles, pole pieces **19** and supports **13** in further embodiments, for example six, eight or more.

[0093] The tube **3** has a low magnetic conductivity and is magnetically insulating in particular. Further, the tube **3** is electrically insulating in order to electrically insulate the pole pieces **19** and the supports **13** from one another. Moreover, the pole pieces **19** are electrically conductive. As a result, a high electric potential can be applied to the pole pieces **19**, while the supports **13** are kept at earth potential. Consequently, the pole pieces **19** in the electric-magnetic multipole element **1**A also serve as electrodes for creating the electric multipole field. However, this also means that the pole pieces **19** are retained in electrically insulated fashion in the mount **27** of the multipole element **1**A; in this case, the material of the mount **27** is electrically nonconductive (at least in part).

[0094] The wall thickness TR of the tube **3** can be sufficiently large to maintain an electrical insulation of the pole pieces **19** vis-à-vis the supports **13** even in the presence of (in terms of absolute value) a high electrical potential difference therebetween. For example, the wall thickness TR of the tube **3** can be dimensioned such that the tube **3** has a breakdown voltage of at least 20 kV. In this case, the breakdown voltage denotes the smallest difference in terms of absolute value between the electric potential of the pole pieces **19** and the electric potential of the supports **13** at which the electrical insulation of the tube **3** fails.

[0095] To apply externally created electric potentials to the pole pieces **19**, the multipole element **1**A also comprises a plurality of first electrical lines **29**, which electrically connect the pole pieces **19** to electrical connectors **31** in the external space assembly **7**. Two first electrical lines **29** are depicted by way of example in FIG. **11**. A region **33** of the tube **3** is emphasized by hatching in FIG. **11**. The region **33** of the tube **3** adjoins the pole pieces **19** and the supports **13** in the radial direction R. In other words, the region **33** of the tube **3** is a region opposite the pole pieces **19** and the supports **13** in the radial direction R. In the region **33**, the tube **3** has no opening (for a vacuum feedthrough) which passes through the tube **3** and the support **13** and the circumferential pole piece **11** in the radial direction R. In contrast to the prior art, which provides for an opening in the region **33** and consequently involves a vacuum seal, the configuration of the present embodiment does not require an opening in the region **33**, facilitating the construction of the tube **3**. By contrast, the tube **3** comprises a vacuum feedthrough **37** away from the region **33**, the vacuum feedthrough maintaining the vacuum in the vacuum space in the radial direction R. The first electrical lines **29** pass through the tube **3** by way of the vacuum feedthrough **37**. That is to say, the first electrical lines **29** pass through the tube **3** in the radial direction R outside of the region **33**. Alternatively, the first lines **29** can be guided through the upper opening **6***a* in the tube **3** or the lower opening **6***b* in the tube **3**. In a further alternative, some of the first electrical lines **29** can be guided through the vacuum feedthrough **37** outside of the region **33** and other first electrical lines **29** can be guided through the upper opening **6***a* in the tube **3** or the lower opening **6***b* in the tube **3**.

[0096] In the examples shown in FIGS. **2** and **11**, the vacuum feedthrough **37** is at a position located between two pole pieces **19** in the circumferential direction U. A possible position of the vacuum feedthrough **37** in the longitudinal direction is depicted in FIG. **2**. From there, it is easy to contact an adjacent pole piece **19**. In addition or in an alternative, a vacuum feedthrough **37** can be arranged above or below a pole piece **19** in the longitudinal direction. The number of vacuum feedthroughs **37** is not limited to one. That is to say, the multipole element **1**A may comprise a plurality of vacuum feedthroughs **37** for guiding the first lines **29** therethrough.

[0097] In the example shown in FIG. **11**, the multipole element **1**A comprises the two first lines **29**, which are electrically connected to two different pole pieces **19**. In order to also be able to apply an external electric potential to the two other pole pieces **19** of the multipole element **1**A, the multipole element **1**A also comprises a plurality of second electrical lines **35** which electrically interconnect at least some of the pole pieces **19** in groups. Specifically, the multipole element **1**A comprises a second electrical line **35** which electrically connects the pole piece **19** arranged at the top in FIG. **11** to the pole piece opposite thereto across the central axis **5** (i.e. the pole piece **19** arranged at the bottom in FIG. **11**). Moreover, the multipole element **1**A comprises a further second electrical line **35** which electrically connects the pole piece **19** arranged to the right in FIG. **11** to the pole piece opposite thereto across the central axis **5** (i.e. the pole piece **19** arranged to the left in FIG. **11**). Hence, each pair of pole pieces **19** opposite one another across the central axis **5** are directly electrically connected to one another. In this example, each pair of pole pieces **19** opposite one another across the central axis **5** forms a group within the sense of the electrical connection "in groups". Such a configuration reduces a negative influence of unwanted fluctuations in the external electric potentials. Like in the example of FIG. **11**, the second lines **35** can be parts of the vacuum space assembly. Alternatively, the second lines **35** can be parts of the external space assembly. However, it is also possible within the scope of a further embodiment that the pole pieces **19** are individually supplied with externally created electric potentials via the first electrical lines **29**.

[0098] During operation, the coils **17**, the supports **13** and the circumferential pole piece **11** can remain at earth potential, whereas the pole pieces **19** are at a high voltage. An electrical potential difference between a mean electric potential of the pole pieces **19** (i.e. an (arithmetic) mean value of the electric potentials of the pole pieces **19**) and an electric potential of the supports **13** is (in terms of absolute value) for example at least 2 kV, such as at least 6 kV. For the purpose of creating an electric quadrupole field with regard to the multipole element **1**A shown in FIG. **11**, an electric potential of for example +2 kV, +4 kV, +2 kV and +4 kV (when counting in the circumferential direction) can be applied to the four pole pieces **19**, while an electric potential of 0 V (i.e. earth potential or ground potential) is applied to the supports **13**. As a result, the mean electric potential of the pole pieces **19** has a value of +3 kV, and an electrical potential difference between the mean electric potential of the pole pieces **19** and an electric potential of the supports **13** is also +3 kV.

[0099] Further or alternatively, an electrical potential difference between a mean electric potential of the pole pieces **19** (i.e. an (arithmetic) mean value of the electric potentials of the pole pieces **19**) and an electric potential of the circumferential pole piece **11** is (in terms of absolute value) at least 2 kV, such as at least 6 kV. Further or alternatively, a maximum electrical potential difference between electric potentials of adjacent pole pieces **19** is (in terms of absolute value) at least 200 V, such as at least 1 kV. Further or alternatively, an electrical potential difference between a mean electric potential of the pole pieces **19** (i.e. an (arithmetic) mean value of the electric potentials of the pole pieces **19**) and an earth potential is (in terms of absolute value) at least 2 kV, such as at least 6 kV. The earth potential can also be referred to as ground potential.

[0100] An exemplary particle beam system **100** is described below with reference to FIG. **12**. For example, the particle beam system **100** can be an electron beam microscope or a processing system having a focused ion beam.

[0101] For example, the particle beam system **100** comprises a particle beam column. The particle beam column comprises a particle source **101** for providing charged particles of a particle beam **102**, for example electrons or ions; an acceleration electrode **103** for accelerating the particles in the particle beam **102**; a beam tube **104**, through which the particle beam **102** runs within the particle beam system **100**; an image error corrector **105**, which comprises at least one of the multipole elements **1**, **1**A described herein; deflection coils **106** and/or deflection electrodes **107** for deflecting the particle beam **102**; and an objective lens **109** for focusing the particle beam **102** on a focal plane FP. The particle beam column comprises a housing in which the aforementioned components are accommodated. For example, the beam tube **104** can extend from the acceleration electrode **103** to the objective lens **109**. The beam tube **104** may be interrupted, for example by the image error corrector **105** (the multipole element **1**). Some components of the particle beam system **100**, for example the image error corrector **105** (the multipole element **1**) and the deflection electrodes **107**, may be arranged, in full or in part, within the beam tube **104**.

[0102] The particle beam system **100** also comprises a vacuum chamber **111**, in which a sample stage **113** is arranged. The sample stage **113** is configured to carry a sample **115**. The sample stage **113** may be configured to displace and rotate the sample **115**.

[0103] The particle beam system **100** also comprises a detector **117** for detecting interaction products **118** created by the interaction of the particle beam **102** with the sample **115**. Interaction products can be for example: secondary electrons, backscattered electrons, secondary ions, backscattered ions, light, radiation, etc.

[0104] The particle beam system **100** also comprises a controller **125** for control and data processing. The controller **125** controls the particle source **101**, the acceleration electrode **103**, the image error corrector **105**, the deflection electrodes **107** and the objective lens **109** via a communications line **121**. The controller **125** controls the sample stage **113** via a communications line **114**. The controller **125** controls the detector **117** and receives data from the detector **117** via a communications line **119**. An input device **127** is connected to the controller **125** in order to receive inputs from a user or a data medium. An output device **129** is connected to the controller **125** in order to output outputs from the controller **125** to a user or a data medium. The controller **125** is connected to a data memory **131** for storing data.

[0105] The particle beam system **100** also comprises one or more current sources and/or one or more voltage sources, which are not depicted in the figures. The current sources and voltage sources are configured to supply the components of the particle beam system **100**, in particular electrodes for creating electric fields and coils for creating magnetic fields, with suitable voltages and electric currents. The current sources and voltage sources are controlled by the controller **125**.

[0106] The beam tube **104** can be at a high voltage during operation. In particular, the same high voltage as applied to the pole pieces **19** can also be applied to the beam tube **104**. For example, this means that the mean electric potential of the pole pieces **19** and the electric potential of the beam tube **104** are (virtually) the same. For example, a difference between the mean electric potential of the pole pieces **19** and the electric potential of the beam tube **104** is (in terms of absolute value) no more than 500 V, such as no more than 100 V. For example, an electrical potential difference between the beam tube **104** and the sample **115** is (in terms of absolute value) at least 2 kV or at least 6 kV. Moreover, a difference between the mean electric potential of the pole pieces **19** and the electric potential of the sample **115** can be (in terms of absolute value) at least 2 kV or at least 6 kV.

[0107] The housing of the particle beam column, the vacuum chamber **111**, the sample **115** or a power supply of the particle beam system **100** can be at earth potential and/or be earthed.

[0108] FIG. **13** shows a schematic cross-sectional view at a connecting point between the beam tube **104** and a multipole element **1** according to an embodiment that differs from the embodiment in FIG. **2**. FIG. **13** shows only an upper portion of some of the components of the multipole element **1**, specifically two pole pieces **19** arranged symmetrically about the central axis **5**, the tube **3** and the upper termination pole piece **12***a*. The lower portion of the multipole element **1** can be formed similarly and can be connected to a lower portion of the beam tube **104**.

[0109] FIG. **13** serves to explain an option for a vacuum-tight connection between the beam tube **104** and the multipole element **1**. The beam tube **104** comprises an end portion **104***a* which provides a flange. The tube **3** comprises an end portion **61** which provides a flange that fits to the flange of the end portion **104***a*. A seal **60** sealing the beam tube **104** and the tube **3** is arranged between the beam tube **104** and the tube **3**, more precisely between the end portion **104***a* and the end portion **61**. For example, the seal **60** is an O-ring.

[0110] For example, a high voltage of +8 kV vis-à-vis earth potential is applied to the beam tube **104**. The upper termination pole piece **12***a* is in contact with the beam tube **104**.

[0111] Therefore, the high voltage of +8 kV vis-à-vis earth potential is also applied to the upper termination pole piece **12***a*. An electrical insulation **62** which electrically insulates the upper termination pole piece **12***a* and the pole pieces **19** from one another is arranged between the upper termination pole piece **12***a* and the pole pieces **19**. As a result, electric potentials that are different from one another and different from the electric potential of the upper termination pole piece **12***a* can be applied to the pole pieces **19**. The electrical

insulation **62** can be formed by an electrically insulating solid material or an empty space (vacuum).

LIST OF REFERENCE SIGNS

[0112] **1**, **1**A Multipole element;
[0113] **3** Tube;
[0114] **4** Propagation region;
[0115] **5** Central axis;
[0116] **6***a*, **6***b* Opening in the tube at the central axis;
[0117] **7** External space assembly;
[0118] **9** Vacuum space assembly;
[0119] **11** Circumferential pole piece;
[0120] **12***a* Upper termination pole piece;
[0121] **12***b* Lower termination pole piece;
[0122] **13** Supports;
[0123] **15** Outer wall surface of the tube;
[0124] **16** Inner wall surface of the tube;
[0125] **17** Coils;
[0126] **19**, **19**A, **19**B, **19**C, **19**D Pole pieces;
[0127] **21**, **22** Lateral surfaces of a pole piece;
[0128] **23** Inner surface of a pole piece;
[0129] **25** Outer surface of a pole piece;
[0130] **27** Mount;
[0131] **29** First electrical line;
[0132] **31** Electrical connectors;
[0133] **33** Region of the tube between pole piece and support;
[0134] **35** Second electrical line;
[0135] **37** Vacuum feedthrough;
[0136] **41** Octupole winding;
[0137] **42** Connectors of the octupole winding;
[0138] **43** First quadrupole winding;
[0139] **44** Connectors of the first quadrupole winding;
[0140] **45** Second quadrupole winding;
[0141] **46** Connectors of the second quadrupole winding;
[0142] **47**, **49** Dipole winding;
[0143] **48**, **50** Connectors of the dipole windings;
[0144] **60** Seal;
[0145] **61** End portion;
[0146] **62** Insulation;
[0147] **100** Particle beam system;
[0148] **101** Particle source;
[0149] **102** Particle beam;
[0150] **103** Acceleration electrode;
[0151] **104** Beam tube;
[0152] **104***a* End portion;
[0153] **105** Image error corrector;
[0154] **106** Deflection coils;
[0155] **107** Deflection electrodes;
[0156] **109** Objective lens;
[0157] **111** Vacuum chamber;
[0158] **113** Sample stage;
[0159] **114** Communications link;
[0160] **115** Sample;
[0161] **117** Detector;
[0162] **118** Interaction products;
[0163] **119** Communications link;
[0164] **121** Communications link;
[0165] **125** Controller;
[0166] **127** Input device;
[0167] **129** Output device;
[0168] **131** Data memory;
[0169] D**1** Distance in the radial direction between pole piece and tube;
[0170] DR Internal diameter of the tube;
[0171] FP Focal plane;
[0172] h**1** First half-line;
[0173] h**2** Second half-line;
[0174] h**3** Third half-line;
[0175] h**4** Fourth half-line;
[0176] k**1** First edge;
[0177] k**2** Second edge;
[0178] k**3** Third edge;
[0179] k**4** Fourth edge;
[0180] L Length of a pole piece in the radial direction;
[0181] R Radial direction;
[0182] R**0** Smallest distance between central axis and pole piece;
[0183] S**1**, S**2** Point of intersection;
[0184] TR Wall thickness of the tube in the radial direction;
[0185] TP Thickness of the pole piece in the circumferential direction;
[0186] U Circumferential direction;
[0187] α Inner coverage angle;
[0188] β Outer coverage angle;
[0189] Φ Coverage angle.

What is claimed is:

1. A multipole element, comprising:

a tube surrounding a central axis of the multipole element; and

an external space assembly outside the tube relative to a radial direction perpendicular to the central axis, the external space assembly outside a vacuum space assembly within the tube relative to the radial direction,

wherein the external space assembly comprises:

a magnetically conductive circumferential pole piece surrounding the tube in a circumferential direction, the circumferential direction being perpendicular to the central axis and to the radial direction;

a plurality of magnetically conductive supports distributed around the central axis and extending from the circumferential pole piece against the radial direction up to an outer wall surface of the tube; and

a plurality of coils,

wherein the vacuum space assembly comprises a plurality of magnetically conductive pole pieces distributed around the central axis and extending from the tube against the radial direction, and

wherein the multipole element configured so that, during use:

an electrical potential difference between a mean electric potential of the pole pieces and an electric potential of the supports is at least 2 kV; and/or

an electrical potential difference between a mean electric potential of the pole pieces and an electric potential of the circumferential pole piece is at least 2 kV.

2. The multipole element of claim **1**, wherein a distance in the radial direction between the tube and any one of the supports is at least zero.

3. The multipole element of claim **1**, wherein a distance in the radial direction between the tube and any one of the supports is less than 20% of an internal diameter of the tube

in the radial direction, and/or a distance in the radial direction between the tube and any one of the supports is less than 5 mm.

4. The multipole element of claim **1**, wherein a distance in the radial direction between the tube and any one of the supports is at least zero.

5. The multipole element of claim **1**, wherein a distance in the radial direction between any one of the pole pieces and the tube is less than 20% of an internal diameter of the tube in the radial direction, and/or a distance in the radial direction between any one of the pole pieces and the tube is less than 5 mm.

6. The multipole element of claim **1**, wherein a wall thickness of the tube in the radial direction is greater than zero.

7. The multipole element of claim **1**, wherein a wall thickness of the tube in the radial direction is less than 20% of an internal diameter of the tube in the radial direction, and/or a wall thickness of the tube in the radial direction is less than 5 mm.

8. The multipole element of claim **1**, wherein the supports do not pass through the tube in the radial direction, and the pole pieces do not pass through the tube in the radial direction.

9. The multipole element of claim **1**, wherein the tube is configured to provide a vacuum separation in the radial direction between the vacuum space assembly and the external space assembly.

10. The multipole element of claim **1**, wherein the tube is magnetically nonconductive.

11. The multipole element of claim **1**, wherein:

each pole piece is configured such that its an inner coverage angle and its an outer coverage angle have different values from one another;

a radius-dependent coverage angle specifies an acute angle between a first half-line and a second half-line;

the first half-line lies in a cross-sectional plane oriented perpendicularly to the central axis, emanates from the central axis, and runs through a first point of intersection;

the second half-line lies in the cross-sectional plane, emanates from the central axis, and runs through a second point of intersection;

the first point of intersection and the second point of intersection are points of intersection of the surface of the respective pole piece with a circle which lies in the cross-sectional plane, has a radius r, and has its center on the central axis;

the inner coverage angle is defined as the largest radius-dependent coverage angle in a range $R0 \leq r < R0 + 20\%\ L$;

the outer coverage angle is defined as the largest radius-dependent coverage angle in a range $DR/2 - 20\%\ L < r \leq DR/2$;

R**0** is the smallest distance in the radial direction between the surface of the respective pole piece and the central axis;

DR is the internal diameter of the tube; and

L is the maximum length of the respective pole piece in the radial direction.

12. The multipole element of claim **11**, wherein:

a sector width is defined as the ratio of 360° to the number of pole pieces; and

each pole piece is shaped such that at least one of the following holds:

its inner coverage angle is greater than its outer coverage angle;

a difference between the inner coverage angle and the outer coverage angle is at least 10% of the sector width;

a ratio of the inner coverage angle to the sector width has a value ranging from 75% to 95%; and

a ratio of the outer coverage angle to the sector width has a value ranging from 35% to 75%.

13. The multipole element of claim **11**, wherein:

each pole piece is shaped such that a mid coverage angle of pole piece is both smaller than the inner coverage angle of the pole piece and smaller than the outer coverage angle of the pole piece (**19**); and

the mid coverage angle is defined as the smallest radius-dependent coverage angle in a range $R0 + 20\%\ L < r < DR/2 - 20\%\ L$.

14. The multipole element of claim **1**, wherein:

each pole piece is shaped such that a mid coverage angle of the pole piece is both smaller than an inner coverage angle of the pole piece and smaller than an outer coverage angle of the pole piece;

a radius-dependent coverage angle specifies an acute angle between a first half-line and a second half-line;

the first half-line lies in a cross-sectional plane oriented perpendicularly to the central axis, emanates from the central axis, and runs through a first point of intersection;

the second half-line lies in the cross-sectional plane, emanates from the central axis, and runs through a second point of intersection;

the first point of intersection and the second point of intersection are points of intersection of the surface of the respective pole piece with a circle which lies in the cross-sectional plane, has the radius r and has its centre on the central axis;

the inner coverage angle is defined as the largest radius-dependent coverage angle in a range $R0 \leq r < R0 + 20\%\ L$;

the outer coverage angle is defined as the largest radius-dependent coverage angle in a range $DR/2 - 20\%\ L < r \leq DR/2$;

the mid coverage angle is defined as the smallest radius-dependent coverage angle in a range $R0 + 20\%\ L < r < DR/2 - 20\%\ L$;

R**0** is the smallest distance in the radial direction between the surface of the respective pole piece and the central axis;

DR is the internal diameter of the tube; and

L is the maximum length of the respective pole piece in the radial direction.

15. The multipole element of claim **1**, wherein:

each pole piece comprises two lateral surfaces, each of which faces an adjacent pole piece of the pole pieces in the circumferential direction; and

the lateral surfaces of the pole pieces are curved or bent.

16. The multipole element of claim **1**, wherein the vacuum space assembly further comprises a mount configured to retain the pole pieces.

17. The multipole element of claim **16**, wherein the tube at least partially comprises the mount.

18. The multipole element of claim **1**, wherein:

the tube is electrically insulating;

the pole pieces are electrically conductive; and

the multipole element further comprises a plurality of first electrical lines electrically connecting the pole pieces to electrical connectors in the external space assembly.

19. The multipole element of claim **18**, wherein at least some of the plurality of first electrical lines extend through an opening in the tube located on the central axis.

20. The multipole element of claim **18**, wherein the plurality of first electrical lines pass through the tube in the radial direction outside a region of the tube adjoining the pole pieces and the supports in the radial direction.

21. The multipole element of claim **18**, further comprising a plurality of second electrical lines electrically interconnecting at least some of the pole pieces in groups.

22. The multipole element of claim **21**, wherein each second electrical line electrically interconnects a pair of pole pieces lying opposite one another across the central axis.

23. The multipole element of claim **1**, wherein a region of the tube adjoining the pole pieces and the supports in the radial direction contains no opening extending through the tube in the radial direction.

24. The multipole element of claim **1**, wherein the multipole element is configured so that, during use, an electrical potential difference between a mean electric potential of the pole pieces and an earth potential is at least 2 kV.

25. The multipole element of claim **1**, wherein:

the multipole element is a quadrupole element comprising four pole pieces and four supports;

the coils comprise at least one winding from a group of windings; and

the group of windings comprises:

a quadrupole field winding configured to create a magnetic quadrupole field; and

a plurality of separated dipole field windings configured to create magnetic dipole fields.

26. The multipole element of claim **1**, wherein:

the multipole element is an octupole element comprising eight pole pieces and eight supports;

the coils comprise at least one winding from a group of windings; and

the group of windings comprises:

an octupole field winding configured to create a magnetic octupole field;

a plurality of separated quadrupole field windings configured to create magnetic quadrupole fields; and

a plurality of separated dipole field windings configured to create magnetic dipole fields.

27. An image error corrector comprising a multipole element according to claim **1**.

28. A particle beam system comprising a multiple element according to claim **1**.

29. The particle beam system of claim **28**, further comprising:

a beam tube; and

a voltage source configured to create an electrical potential difference of at least 2 kV between the beam tube and a sample.

* * * * *